EXPONENTIAL SUMS OVER ORDINATES OF NONTRIVIAL ZEROS OF THE RIEMANN ZETA FUNCTION

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Abstract. We study exponential sums formed from the ordinates of the nontrivial zeros of the Riemann zeta function through the geometry of their centered phase walk. Landau's fixed-real formula at $X = e$ gives a logarithmic confinement baseline for the complete-zero walk $Z_N = \sum_{k \le N} e^{\rho_k - \frac{1}{2}}$, and under the Riemann Hypothesis (RH) this specializes to the unit phase walk $S_N = \sum_{k \le N} e^{i\gamma_k}$. The main structural point is that centering forces an exact Brownian-bridge covariance kernel, equivalently the Green kernel of the finite Dirichlet Laplacian. This turns the centered moment into a weighted pair-correlation statistic and gives a sine-mode Green decomposition. The same modal language also expresses the Montgomery–GUE form factor as a constant-mode projection of the Green-modal vector.

Contents

1. Introduction 2
Notation 3
Notation 3
Phase conventions 4
2. The Riemann Zeta Function 4
2.1. Definition 4
2.2. Analytic Continuation 5
2.3. Functional Equation 5
2.4. Special Values and Basic Identities 5
2.5. The Zeros of the Zeta Function 5
3. The Riemann Hypothesis 6
4. Uniform Distribution of Zeta Ordinates 6
4.1. The Landau–Gonek Formula 7
4.2. Chebyshev function in logarithmic frequency 8
5. The Exponential Sum S_N 10
5.1. Definition of the Phase Walk 10
6. A Logarithmic Quadratic Bound for the Centered Phase Walk 11
6.1. Specialization under the Riemann Hypothesis 16
6.2. Roadmap of the logarithmic bound 17
6.3. Pair-correlation form 17
6.4. The Brownian-bridge Green kernel forced by centering 20
6.5. Spectral lifting and the Green-kernel modal decomposition 21
6.6. Green-generated Fourier pair statistics 26
6.7. Endpoint projection and Green deweighting 28
6.8. Unconditional Green deweighting of the endpoint projection 32
6.9. The Green-modal form factor and the Montgomery–GUE target 37
6.10. From the deformed phase walk to the sine kernel 38
6.11. Finite modal identities and the scaled modal variable 40

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6.12. Smoothed modal covariance and conditional form-factor criteria 42
6.13. Riemann–von Mangoldt background and modal residuals 46
7. Supplementary Exact Product Forms for S_N and C_N 48
7.1. A Finite Product Identity 48
7.2. The Phase Product and the Sum S_N 49
7.3. Polynomial Form and Elementary Symmetric Coefficients 50
7.4. Product Form of the Centroid 50
7.5. The Triangular Product 51
7.6. Triangular Harmonic Interpretation 51
7.7. Local Analytic Interpretation at the Origin 52
8. Supplementary Non-Resonant Triangular Landau–Gonek Structure of S_N and C_N 52
8.1. Ordinal and height truncations 52
8.2. Continuous phase sums 53
8.3. Arithmetic Landau–Gonek resonances 54
8.4. The centroid as a triangular non-resonant transform 54
8.5. Equidistribution consequences 55
8.6. Structural conclusion 57
9. Conclusions 57
9.1. Complete-zero logarithmic confinement 57
9.2. Exact centered pair statistic 57
9.3. Brownian-bridge/Green kernel 58
9.4. Lifted phases and Green-modal coordinates 58
9.5. Exact recovery versus scalar energy bounds 58
9.6. Green-modal Montgomery formulation 59
9.7. RVM residual decomposition 59
9.8. Structural separation 60
References 60

1. Introduction

The nontrivial zeros of the Riemann zeta function are written $\rho = \beta + i\gamma$. Their ordinates γ enter both the Riemann–von Mangoldt counting formula and the fine-scale pair-correlation problem. The present paper studies these ordinates through the phase walk obtained from the increments $e^{i\gamma_k}$. The basic objects are the endpoint sum S_N, its triangular centroid C_N, and the centered quadratic moment $M_0(N)$.

The preliminary sections recall the standard zeta-function background: the definition and analytic continuation of $\zeta(s)$, the functional equation, the critical strip, the Riemann Hypothesis, Weyl equidistribution, Hlawka's theorem for zeta ordinates, and the Landau–Gonek distinction between resonant prime-power frequencies and non-resonant continuous frequencies. These are included to fix notation and to separate classical analytic estimates from the structural observation made here.

The core construction begins with the complete-zero auxiliary walk

$$Z_N = \sum_{k=1}^{N} e^{\rho_k - \frac{1}{2}}, \tag{1.1}$$

with centroid $\mathcal{C}_N$ and centered moment $\mathcal{M}_0(N)$. Landau's fixed-real formula at $X = e$ supplies the logarithmic confinement baseline. The new observation is that the centered phase-walk moment

realizes this confinement through an exact Brownian-bridge/Green-kernel pair statistic, and that the same Green-modal language gives a natural expression of the Montgomery–GUE form factor.

Thus the logical chain is

(1.2)
$$\begin{aligned}\text{centered phase-walk geometry} &\Longrightarrow \text{Brownian-bridge/Green kernel}\\ &\Longrightarrow \text{weighted pair correlation}\\ &\Longrightarrow \text{Green-modal expression of the Montgomery–GUE form factor.}\end{aligned}$$

A central point of the paper is the separation between the fixed phase walk and the deformed phase-walk family. The original walk $S_N = S_N(1)$ is the non-resonant Landau section associated with $X = e$. The Montgomery–GUE statistics are obtained from the scaled endpoint $S_N(t)$, with $t = \tau \log(T/2\pi)$. The Green-modal framework places the fixed-frequency geometry and the Montgomery endpoint form factor in a common coordinate system.

The Montgomery–GUE material below is formulated through exact finite Green-modal identities, formal equivalences, and the corresponding form-factor target.

Notation

$\rho_k = \beta_k + i\gamma_k$	nontrivial zeros with positive ordinate, counted with multiplicity
γ_k	ordinate of a zero, measured in radians
$e^{i\gamma_k}$	unit phase associated with γ_k
$e^{it\gamma_k}$	t-deformed ordinate phase
$B_k,\ \theta_k$	lifted winding number and angular representative of γ_k
$e^{2\pi i x}$	Weyl/modulo-one phase convention
$e^{\rho_k - \frac{1}{2}}$	complete-zero phase; equals $e^{i\gamma_k}$ under RH
$S_N,\ C_N,\ M_0(N)$	unit-phase endpoint, triangular centroid, and centered moment
$Z_N,\ \mathcal{C}_N,\ \mathcal{M}_0(N)$	complete-zero endpoint, centroid, and centered moment
$W_N(j,k)$	centering weight in the original ordinal variables
$r = N-j+1,\ s = N-k+1$	reversed pair indices for $W_N(j,k)$
$r = N-k+1$	reversed single index for endpoint/modal vectors
$K_N(r,s)$	reversed Brownian-bridge/Green kernel
$\phi_m(r),\ \lambda_m$	Dirichlet sine modes and Green eigenvalues
$A_m(N;t),\ \mathcal{A}_m(N)$	unit-phase t-modal coefficient and complete-zero modal coefficient
u_m	projection of the constant vector onto the m-th sine mode
$v_r,\ w_m$	centroid weight vector and its sine-mode coefficients
$T_N,\ T_0,\ L_N$	$T_N = \gamma_N$; T_0 is a representative local-window height; $L_N = \log(T_N/2\pi)/(2\pi)$
t	Fourier parameter; smoothed Montgomery scaling: $t = \tau\log(T_N/2\pi) + y$
$\mathfrak{F}_N(\tau),\ F_N(\tau)$	unsmoothed and smoothed Montgomery form factors
ξ_{jk}	unfolded pair variable $\xi_{jk} = \frac{\log(T_0/2\pi)}{2\pi}(\gamma_j - \gamma_k)$
$\Sigma_N(\tau)$	smoothed modal covariance matrix
$N_0(T),\ \widetilde{\gamma}_k$	smooth Riemann–von Mangoldt main term and its inverse grid
$d_k,\ R_m(N;t)$	zero displacement from the RVM grid and the corresponding modal residual

Phase conventions. Throughout the paper, ordinates γ_k are real numbers measured in radians. The basic unit phase is therefore

$$e^{i\gamma_k}. \tag{1.3}$$

For a real Fourier parameter t, the corresponding t-deformed phase is

$$e^{it\gamma_k}. \tag{1.4}$$

When uniform distribution modulo one is discussed, the standard Weyl notation is used:

$$e^{2\pi i m x_n}. \tag{1.5}$$

Thus, if one writes $x_n = \alpha\gamma_n$ modulo one, Weyl sums have the form

$$e^{2\pi i m \alpha \gamma_n}. \tag{1.6}$$

Taking $\alpha = 1/(2\pi)$ and $m = 1$ gives exactly

$$e^{i\gamma_n}. \tag{1.7}$$

The complete-zero phase is a different object:

$$e^{\rho_k - \frac{1}{2}}, \qquad \rho_k = \beta_k + i\gamma_k. \tag{1.8}$$

Under RH this specializes to

$$e^{\rho_k - \frac{1}{2}} = e^{i\gamma_k}. \tag{1.9}$$

Thus the paper uses $e^{i\gamma_k}$ for radian phases, $e^{2\pi i x}$ only for modulo-one Weyl notation, $e^{it\gamma_k}$ for Fourier deformation, and $e^{\rho_k - \frac{1}{2}}$ for complete-zero expressions.

2. The Riemann Zeta Function

This section contains standard material. The classical references are [2], [3], and [4].

2.1. **Definition.**

Definition 2.1. The Riemann zeta function is defined for $\Re(s) > 1$ by the absolutely convergent series

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}. \tag{2.1}$$

Euler showed that the same function admits the product expansion

$$\zeta(s) = \prod_{p} \left(1 - p^{-s}\right)^{-1}, \qquad \Re(s) > 1, \tag{2.2}$$

where the product extends over all prime numbers. Equation (2.2) establishes the fundamental connection between $\zeta(s)$ and the distribution of primes.

2.2. **Analytic Continuation.** A classical theorem of Riemann states that $\zeta(s)$ admits a meromorphic continuation to the whole complex plane, with a unique simple pole at

$$s = 1. \tag{2.3}$$

See [3] or [2].

2.3. **Functional Equation.** Define

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s). \tag{2.4}$$

The completed zeta function satisfies

$$\xi(s) = \xi(1-s). \tag{2.5}$$

Equivalently,

$$\zeta(s) = 2^s\pi^{\,s-1}\sin\left(\frac{\pi s}{2}\right)\Gamma(1-s)\zeta(1-s). \tag{2.6}$$

The symmetry encoded by (2.5) plays a central role in the study of the zeros of $\zeta(s)$.

2.4. **Special Values and Basic Identities.** Among the classical identities are

$$\zeta(2) = \frac{\pi^2}{6}, \tag{2.7}$$

and more generally

$$\zeta(2m) = (-1)^{m+1}\frac{B_{2m}(2\pi)^{2m}}{2(2m)!}, \qquad m \geq 1, \tag{2.8}$$

where B_{2m} denotes the Bernoulli numbers.
These identities are standard and may be found in [2] or [3].

2.5. **The Zeros of the Zeta Function.** The zeros of $\zeta(s)$ are traditionally divided into two classes.

Definition 2.2. The trivial zeros are located at

$$s = -2, -4, -6, \ldots. \tag{2.9}$$

All remaining zeros are called nontrivial zeros.

Definition 2.3. A nontrivial zero is a complex number

$$\rho = \beta + i\gamma \tag{2.10}$$

such that

$$\zeta(\rho) = 0, \tag{2.11}$$

with

$$0 < \beta < 1. \tag{2.12}$$

The region (2.12) is known as the critical strip.

The functional equation implies that the nontrivial zeros occur in symmetric sets

$$\rho, \quad \overline{\rho}, \quad 1-\rho, \quad 1-\overline{\rho}. \tag{2.13}$$

Definition 2.4. For $T > 0$, let

$$N_\zeta(T) = \#\{\rho = \beta + i\gamma : \zeta(\rho) = 0,\ 0 < \beta < 1,\ 0 < \gamma \le T\}, \tag{2.14}$$

where zeros are counted with multiplicity.

The Riemann–von Mangoldt formula gives the asymptotic estimate

$$N_\zeta(T) = \frac{T}{2\pi}\log\left(\frac{T}{2\pi}\right) - \frac{T}{2\pi} + O(\log T). \tag{2.15}$$

See [2] or [3].

3. The Riemann Hypothesis

This section contains standard material.

Conjecture 3.1 (Riemann Hypothesis)**.** *Every nontrivial zero of the Riemann zeta function satisfies*

$$\Re(\rho) = \frac{1}{2}. \tag{3.1}$$

Assuming (3.1), all nontrivial zeros can be written as

$$\rho_n = \frac{1}{2} + i\gamma_n. \tag{3.2}$$

The sequence of positive ordinates is written as

$$0 < \gamma_1 \le \gamma_2 \le \gamma_3 \le \cdots, \tag{3.3}$$

with multiplicities counted. This sequence is the principal object studied throughout this work.

The Riemann Hypothesis is one of the most important open problems in mathematics and has deep implications for the distribution of prime numbers. Comprehensive discussions may be found in [2], [3], and [4].

4. Uniform Distribution of Zeta Ordinates

The study of exponential sums over the ordinates of the nontrivial zeros is closely related to the theory of uniform distribution modulo one.

The fundamental tool is Weyl's criterion.

Theorem 4.1 (Weyl Criterion)**.** *Let (x_n) be a sequence of real numbers.*

Then (x_n) is uniformly distributed modulo one if and only if

$$\lim_{N\to\infty} \frac{1}{N}\sum_{n=1}^{N} e^{2\pi i m x_n} = 0 \qquad (m \neq 0). \tag{4.1}$$

A natural question is therefore whether the scaled ordinates $\alpha\gamma_n$ are uniformly distributed modulo one. The first systematic investigation of this problem appears in the work of Rademacher [6], who studied the height-truncated sums

$$A_\alpha(T) = \sum_{0<\gamma\le T} e^{2\pi i\alpha\gamma}. \tag{4.2}$$

Subsequent work by Hlawka established the corresponding equidistribution theorem.

Theorem 4.2 (Hlawka, 1975)**.** *For every nonzero real number α, the sequence $\alpha\gamma_n$ is uniformly distributed modulo one. Equivalently,*

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} e^{2\pi i m\alpha\gamma_n} = 0 \qquad (m \neq 0). \tag{4.3}$$

Taking $\alpha = 1/(2\pi)$ and $m = 1$ in (4.3) gives

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} e^{i\gamma_n} = 0. \tag{4.4}$$

This is the classical normalized cancellation of the ordinal phase sums. The consequences for the main objects of the paper are recorded after those objects are defined.

The exceptional frequencies that survive averaging are described by the Landau–Gonek theory discussed in Subsection 4.1.

4.1. **The Landau–Gonek Formula.**

Definition 4.3 (von Mangoldt Function)**.** The von Mangoldt function is defined by

$$\Lambda(\ell) = \begin{cases} \log p, & \ell = p^m, \quad p \text{ prime}, \quad m \ge 1, \\ 0, & \text{otherwise.} \end{cases} \tag{4.5}$$

The function $\Lambda(\ell)$ therefore detects prime powers and assigns to each prime power p^m the weight $\log p$.

The arithmetic weights which appear in the Landau–Gonek formula originate in the logarithmic derivative of the Euler product. For $\Re(s) > 1$, the identity

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{\ell=1}^{\infty} \frac{\Lambda(\ell)}{\ell^s} \tag{4.6}$$

holds. Equivalently, the Chebyshev function

$$\psi(x) = \sum_{\ell\le x} \Lambda(\ell) \tag{4.7}$$

is the von Mangoldt weighted summatory function naturally coupled to the zeros through the explicit formula. Thus the same von Mangoldt weights which occur in the logarithmic derivative also determine the arithmetic frequencies that appear in Landau–Gonek sums over zero ordinates.

4.2. **Chebyshev function in logarithmic frequency.** Let

$$\psi(x) = \sum_{n \le x} \Lambda(n) \tag{4.8}$$

be the Chebyshev function. In the logarithmic variable

$$u = \log x, \tag{4.9}$$

write

$$\Psi(u) = \psi(e^u). \tag{4.10}$$

The classical explicit formula may be written schematically, away from prime-power discontinuities and with the usual interpretation of the zero sum, as

$$\Psi(u) = e^u - \sum_{\rho} \frac{e^{\rho u}}{\rho} - \log(2\pi) - \frac{1}{2}\log(1 - e^{-2u}). \tag{4.11}$$

Thus the zero contribution to $\Psi(u)$ is built from the exponential modes

$$e^{\rho u}. \tag{4.12}$$

Formally differentiating with respect to u gives

$$\frac{d}{du}\Psi(u) = e^u - \sum_{\rho} e^{\rho u} - \frac{e^{-2u}}{1 - e^{-2u}}, \tag{4.13}$$

again in the regularized or distributional sense appropriate to the explicit formula.
Consequently, the spectral quantity

$$\sum_{\rho} e^{\rho u} \tag{4.14}$$

is the logarithmic-frequency zero side associated with the Chebyshev function. The special value $u = 1$ corresponds to the continuous frequency

$$X = e^u = e. \tag{4.15}$$

At this point one obtains the complete-zero phase

$$e^{\rho - \frac{1}{2}} = e^{-\frac{1}{2}} e^{\rho}. \tag{4.16}$$

Under RH,

$$\rho = \frac{1}{2} + i\gamma, \tag{4.17}$$

and hence

$$e^{\rho - \frac{1}{2}} = e^{i\gamma}. \tag{4.18}$$

Thus the ordinate phase sum

$$\sum_{0<\gamma\le T} e^{i\gamma} \tag{4.19}$$

is, under RH, the $u = 1$ specialization of the logarithmic-frequency zero side of the explicit formula.

This passage should not be read as an ordinary pointwise Fourier inversion from $\psi(x)$ to S_N. The explicit formula is a regularized identity. Height truncation, smoothing, or distributional interpretation is needed to make the zero sum precise. The ordinal sum S_N is then related to the height-truncated sum through the height-to-ordinal bridge.

This regularized Chebyshev discussion is separate from the later application of Landau's fixed-real formula at $X = e$. The Landau input used for the logarithmic bound is the height-truncated fixed-X estimate in Proposition 6.2; it does not use a symmetric truncation, smoothing in height, or a distributional interpretation at $u = 1$.

In this regularized logarithmic-frequency sense, the height-truncated phase sums

$$S_x(T) = \sum_{0<\gamma\le T} x^{i\gamma}, \qquad x \in \mathbb{N},\ x > 1, \tag{4.20}$$

may be viewed as logarithmic-frequency probes of the zero side of the explicit formula. Prime powers are distinguished because they are precisely the support of Λ.

Unlike the metric theory of Hlawka, the Landau–Gonek formula studies special frequencies associated with prime powers.

For a fixed integer $x > 1$, the behaviour of (4.20) depends strongly on the arithmetic nature of x.

It plays a central role in the explicit formulae relating prime numbers and nontrivial zeros of the Riemann zeta function [8, 2, 4].

The classical Landau formula is naturally stated for the zero sum

$$L_x(T) = \sum_{0<\gamma\le T} x^{\rho}. \tag{4.21}$$

For a fixed integer $x > 1$, Landau's formula gives

$$L_x(T) = -\frac{T}{2\pi}\Lambda(x) + O(\log T). \tag{4.22}$$

In particular, when x is a prime power,

$$x = p^m, \tag{4.23}$$

the von Mangoldt term in (4.22) is nonzero.

Assuming the Riemann Hypothesis, each nontrivial zero has the form

$$\rho = \frac{1}{2} + i\gamma. \tag{4.24}$$

Therefore

$$x^{\rho} = \sqrt{x}\, x^{i\gamma}. \tag{4.25}$$

Thus the passage from Landau's zero sum in x^{ρ} to the pure phase sum in $x^{i\gamma}$ introduces the factor $x^{-1/2}$.

The modern quantitative form of this phenomenon is due to Gonek [10], who refined Landau's original argument using the explicit formula.

Theorem 4.4 (Landau–Gonek, phase form under RH)**.** *For frequencies associated with prime powers, the pure phase sum* (4.20) *admits, under the Riemann Hypothesis, a main term proportional to the von Mangoldt function divided by* $\sqrt{x}$*.*

In the fixed-x form,

$$\sum_{0<\gamma\le T} x^{\,i\gamma} = -\frac{T}{2\pi}\frac{\Lambda(x)}{\sqrt{x}} + O(\log T). \tag{4.26}$$

Consequently, prime powers define distinguished resonant frequencies for exponential sums over the ordinates of the nontrivial zeros.

This phenomenon stands in sharp contrast with the metric equidistribution results discussed earlier in this section.

Remark 4.5. Theorem 4.4 shows that exponential sums over the ordinates of the zeta zeros possess distinguished resonant frequencies indexed by prime powers.

From the viewpoint of Weyl equidistribution, these frequencies are exceptional: they generate a nonzero main term at the Landau–Gonek scale.

The contrast between the generic behaviour described by Theorem 4.2 and the resonant behaviour described by Theorem 4.4 is one of the central themes underlying the study of exponential sums over zeta ordinates.

5. The Exponential Sum S_N

This section introduces the main objects investigated in this paper.

5.1. **Definition of the Phase Walk.** For each ordinate γ_n we associate the unit complex phase

$$z_n = e^{i\gamma_n}. \tag{5.1}$$

Definition 5.1. The cumulative exponential sum is defined by

$$S_N = \sum_{k=1}^{N} e^{i\gamma_k}. \tag{5.2}$$

Equivalently,

$$S_N = \sum_{k=1}^{N} \cos(\gamma_k) + i\sum_{k=1}^{N} \sin(\gamma_k). \tag{5.3}$$

The sequence

$$S_1, S_2, \ldots, S_N \tag{5.4}$$

defines a trajectory in the complex plane obtained by successively adding the unit vectors $e^{i\gamma_k}$.

Definition 5.2. The centroid of the phase-walk trajectory is defined by

$$C_N = \frac{1}{N}\sum_{n=1}^{N} S_n. \tag{5.5}$$

Using the definition of S_n, the centroid can also be written as

$$C_N = \frac{1}{N}\sum_{n=1}^{N}\sum_{k=1}^{n} e^{i\gamma_k}. \tag{5.6}$$

Interchanging the order of summation gives the weighted form

$$C_N = \frac{1}{N}\sum_{k=1}^{N}(N-k+1)e^{i\gamma_k}. \tag{5.7}$$

Thus C_N is not the simple average of the phases. It is the average of the partial sums S_n, or equivalently a triangularly weighted average of the increments $e^{i\gamma_k}$.

The geometric and distributional properties of the trajectory (S_n), together with the behaviour of its centroid C_N, constitute the central subject of the present paper.

Remark 5.3. Classical equidistribution gives normalized cancellation of the phase increments. Its precise consequences for the endpoint S_N and for the triangular centroid C_N are isolated in Section 8.

6. A Logarithmic Quadratic Bound for the Centered Phase Walk

This section separates the height-truncated statement from the ordinal statement and keeps track of possible multiplicities at the boundary height. The natural object before imposing RH is the complete-zero phase $e^{\rho-1/2}$. Under RH this becomes the unit phase $e^{i\gamma}$ used in the original walk S_N.

Definition 6.1 (Fixed-real Landau resonance function)**.** For $X > 1$, define

$$\Lambda_{\mathbb{R}}(X) = \begin{cases} \log p, & X = p^m, \quad p \text{ prime}, \quad m \in \mathbb{N}, \\ 0, & \text{otherwise.} \end{cases} \tag{6.1}$$

This is the fixed-real Landau resonance function. It agrees with the von Mangoldt weight at integral prime powers and is zero away from them. It is not the ordinary von Mangoldt function on the integers, but the fixed-real extension appearing in Landau's formula.

Proposition 6.2 (Landau's fixed-real formula)**.** *Let $X > 1$ be fixed and real, and write*

$$X^\rho = \exp(\rho \log X), \tag{6.2}$$

where $\log X$ denotes the real logarithm. Then, as $T \to \infty$,

$$\sum_{0<\gamma\le T} X^\rho = -\frac{T}{2\pi}\Lambda_{\mathbb{R}}(X) + O_X(\log T), \tag{6.3}$$

where the sum is over the nontrivial zeros $\rho = \beta + i\gamma$ of $\zeta(s)$ with positive ordinate, counted with multiplicity. The implied constant may depend on the fixed value of X. No uniformity in X is used in this paper.

Equivalently, for each fixed $X > 1$ there are constants $C_L(X) > 0$ and $T_L(X) \ge 2$ such that, for all $T \ge T_L(X)$,

$$\left| \sum_{0<\gamma\le T} X^\rho + \frac{T}{2\pi}\Lambda_{\mathbb{R}}(X) \right| \le C_L(X)\log T. \tag{6.4}$$

Proof. This is Landau's fixed-frequency formula. Landau proved the formula for fixed $X > 1$. The fixed-real formulation used here is the form recalled by Kaczorowski, Languasco, and Perelli in the displayed fixed-X formula in the introduction of [12] (J. Kaczorowski, A. Languasco, A. Perelli, *A note on Landau's formula*, Functiones et Approximatio Commentarii Mathematici **28** (2000), 173–186):

$$\sum_{0<\gamma\le T} X^{\rho} = -\frac{T}{2\pi}\Lambda_{\mathbb{R}}(X) + O_X(\log T), \qquad T \to \infty, \tag{6.5}$$

where $\Lambda_{\mathbb{R}}(X) = \log p$ if $X = p^m$, p prime, and $\Lambda_{\mathbb{R}}(X) = 0$ otherwise. Uniform versions in X belong to the Landau–Gonek line of results, beginning with Gonek's work. The present paper uses only the single fixed value $X = e$, so Landau's fixed-X form is sufficient.

The convention (6.2) fixes the meaning of X^{ρ} for real $X > 1$. No smoothing, regularization, or limiting approximation of $X = e$ by integer frequencies is used in this application of the fixed-real formula. □

Remark 6.3 (No approximation by integer frequencies). The use of $X = e$ below is not obtained by approximating e by integers. Such an approximation would require a uniform Landau–Gonek theorem in X, because the factor $X^{i\gamma}$ is sensitive to $\log X$ at height T. Instead, Proposition 6.2 is applied directly at the single fixed real frequency $X = e$.

Remark 6.4 (Arithmetic term and error constants). The Landau formula used here is not an error-free identity. It has an arithmetic main term and an analytic error term. In the notation of Proposition 6.2,

$$\sum_{0<\gamma\le T} X^{\rho} = -\frac{T}{2\pi}\Lambda_{\mathbb{R}}(X) + E_L(T;X), \qquad |E_L(T;X)| \le C_L(X)\log T \tag{6.6}$$

for $T \ge T_L(X)$. At the non-resonant fixed value $X = e$, the arithmetic term vanishes because $\Lambda_{\mathbb{R}}(e) = 0$, and the complete-zero height sum is controlled by the remaining error term:

$$|\mathcal{Z}(T)| \le C_L(e)\log T \qquad (T \ge T_L(e)). \tag{6.7}$$

Thus $C_L(e)$ is the fixed-$X = e$ error constant supplied by the chosen quantitative Landau formulation; it is not an additional identity.

Corollary 6.5 (The non-resonant frequency $X = e$)**.** *At the fixed real frequency $X = e$,*

$$\Lambda_{\mathbb{R}}(e) = 0. \tag{6.8}$$

Consequently,

$$\sum_{0<\gamma\le T} e^{\rho} = O(\log T), \tag{6.9}$$

and therefore

$$\mathcal{Z}(T) := \sum_{0<\gamma\le T} e^{\rho-\frac{1}{2}} = O(\log T). \tag{6.10}$$

Proof. The number e is not an integral prime power. Hence, by Definition 6.1,

$$\Lambda_{\mathbb{R}}(e) = 0. \tag{6.11}$$

Applying Proposition 6.2 with $X = e$ gives

$$\sum_{0<\gamma\le T} e^{\rho} = O(\log T). \tag{6.12}$$

Finally,

$$e^{\rho-\frac{1}{2}} = e^{-1/2}e^{\rho}, \tag{6.13}$$

so

$$\mathcal{Z}(T) = e^{-1/2} \sum_{0<\gamma\le T} e^{\rho} = O(\log T). \tag{6.14}$$

This proves (6.10). □

Remark 6.6 (Height truncation and multiplicities). Corollary 6.5 gives the primary height-truncated estimate. It counts every zero at a given ordinate with its multiplicity. The ordinal theorem below passes from height truncation to Z_N by isolating the boundary block of zeros at the last ordinate.

Definition 6.7 (Ordinal complete-zero walk)**.** Let $\rho_k = \beta_k + i\gamma_k$ denote the nontrivial zeros with positive ordinate, ordered by nondecreasing ordinate and counted with multiplicity. Define

$$Z_N = \sum_{k=1}^{N} e^{\rho_k-\frac{1}{2}}, \tag{6.15}$$

Its centroid is

$$\mathcal{C}_N = \frac{1}{N}\sum_{n=1}^{N} Z_n, \tag{6.16}$$

and its centered quadratic moment is

$$\mathcal{M}_0(N) = \frac{1}{N}\sum_{n=1}^{N} |Z_n - \mathcal{C}_N|^2 . \tag{6.17}$$

Lemma 6.8 (Multiplicity at a fixed ordinate)**.** *Let $m(\Gamma)$ denote the number of nontrivial zeros with positive ordinate equal to Γ, counted with multiplicity. There is an absolute constant $C_{\mathrm{RVM}} > 0$, coming only from the Riemann–von Mangoldt estimate in unit intervals, such that for all $\Gamma \ge 2$,*

$$m(\Gamma) \le C_{\mathrm{RVM}} \log \Gamma. \tag{6.18}$$

Proof. It suffices to consider Γ large; bounded values of Γ are absorbed into the implied constant. By definition,

$$m(\Gamma) \le N_\zeta(\Gamma+1) - N_\zeta(\Gamma-1). \tag{6.19}$$

The Riemann–von Mangoldt formula gives, for some absolute C_{RVM} and all $T \ge 2$,

$$N_\zeta(T+1) - N_\zeta(T-1) \le C_{\mathrm{RVM}} \log T, \tag{6.20}$$

and hence $m(\Gamma) \le C_{\mathrm{RVM}} \log \Gamma$. □

Lemma 6.9 (Height-to-ordinal passage with multiplicities, effective form)**.** *Let the nontrivial zeros with positive ordinate be ordered by nondecreasing ordinate and counted with multiplicity:*

$$\rho_k = \beta_k + i\gamma_k, \qquad 0 < \gamma_1 \le \gamma_2 \le \cdots . \tag{6.21}$$

Define

$$Z_N = \sum_{k=1}^{N} e^{\rho_k - \frac{1}{2}}. \tag{6.22}$$

There are positive constants $C_L(e), C_{\mathrm{RVM}}, C_{\mathrm{ord}}, T_0, N_0$. Put

$$C_Z = C_{\mathrm{ord}} \left(C_L(e) + e^{1/2} C_{\mathrm{RVM}} \right), \tag{6.23}$$

Then, for every $N \ge N_0$,

$$|Z_N| \le C_Z \log N. \tag{6.24}$$

In particular,

$$Z_N = O(\log N). \tag{6.25}$$

Proof. By Landau's fixed-real formula at $X = e$, there are constants $C_L(e) > 0$ and $T_L \ge 2$ such that, for all $T \ge T_L$,

$$|\mathcal{Z}(T)| = \left| \sum_{0 < \Im\rho \le T} e^{\rho - \frac{1}{2}} \right| \le C_L(e) \log T. \tag{6.26}$$

Let

$$\Gamma_N = \gamma_N. \tag{6.27}$$

For this proof write $T_N = \Gamma_N$. Let

$$I_N = \{k : \gamma_k = T_N\} \tag{6.28}$$

be the full multiplicity block at height T_N, and let $J_N \subseteq I_N$ be the part of that block included among the first N zeros. Since the positive ordinates form a discrete multiset, one may choose $T_N^- < T_N$ with no zero ordinate in (T_N^-, T_N). Then the height-truncated sum up to T_N^- contains exactly the first zeros before the boundary block, while B_N contains exactly the selected part J_N of the boundary block:

$$Z_N = \mathcal{Z}(T_N^-) + B_N, \tag{6.29}$$

where

$$B_N = \sum_{k \in J_N} e^{\rho_k - \frac{1}{2}}. \tag{6.30}$$

If $T_N \ge T_0 := \max(T_L, 2)$, then (6.26) gives

$$|\mathcal{Z}(T_N^-)| \le C_L(e) \log T_N. \tag{6.31}$$

Moreover,

$$|B_N| \le \sum_{k\in J_N} \left|e^{\rho_k-\frac{1}{2}}\right| = \sum_{k\in J_N} e^{\beta_k-\frac{1}{2}}. \tag{6.32}$$

Since $0 < \beta_k < 1$,

$$e^{\beta_k-\frac{1}{2}} \le e^{1/2}. \tag{6.33}$$

Thus

$$|B_N| \le e^{1/2} m(T_N) \le e^{1/2}\left(N_\zeta(T_N+1) - N_\zeta(T_N-1)\right) \le e^{1/2} C_{\mathrm{RVM}} \log T_N. \tag{6.34}$$

Here the last estimate is Lemma 6.8, which uses only the Riemann–von Mangoldt formula in a unit interval and does not require simple ordinates. Therefore

$$|Z_N| \le \left(C_L(e) + e^{1/2} C_{\mathrm{RVM}}\right) \log T_N. \tag{6.35}$$

Finally, the Riemann–von Mangoldt formula (2.15) implies $T_N = \gamma_N \ll N \log N$ for N large. Hence there is a constant $C_{\mathrm{ord}} > 0$ such that

$$\log T_N \le C_{\mathrm{ord}} \log N \tag{6.36}$$

for every $N \ge N_0$, after increasing N_0 if necessary so that $T_N \ge T_0$. Substituting this estimate in (6.35) gives (6.24) with C_Z as in (6.23).

The $O(\log N)$ statement follows immediately. □

Remark 6.10 (From asymptotic notation to explicit inequalities). The preceding proof is the exact mechanism by which the asymptotic notation is converted into an explicit inequality. Once constants are fixed, the conversion is mechanical. Landau's fixed-real formula at $X = e$ supplies $C_L(e)$ and T_L such that

$$|\mathcal{Z}(T_N^-)| \le C_L(e) \log T_N \qquad (T_N \ge T_0). \tag{6.37}$$

The boundary block is controlled by Riemann–von Mangoldt in a unit interval:

$$|B_N| \le e^{1/2} m(T_N) \le e^{1/2} C_{\mathrm{RVM}} \log T_N. \tag{6.38}$$

The exact decomposition

$$Z_N = \mathcal{Z}(T_N^-) + B_N \tag{6.39}$$

therefore gives

$$|Z_N| \le \left(C_L(e) + e^{1/2} C_{\mathrm{RVM}}\right) \log T_N \tag{6.40}$$

for all sufficiently large N. Finally, the ordinal height relation $\log T_N \le C_{\mathrm{ord}} \log N$, valid for $N \ge N_0$ after fixing the ordinal threshold, gives

$$|Z_N| \le C_Z \log N, \qquad C_Z = C_{\mathrm{ord}}\left(C_L(e) + e^{1/2} C_{\mathrm{RVM}}\right). \tag{6.41}$$

Thus effective claims require the constants $C_L(e)$, C_{RVM}, C_{ord}, and the threshold N_0 to be fixed. The big-O statement records the same chain without specifying their values. Here $C_L(e)$ comes from

the fixed-$X = e$ Landau error term, C_{RVM} comes from the Riemann–von Mangoldt unit-interval multiplicity bound, and C_{ord} comes from the height-to-ordinal comparison $\log T_N \le C_{\mathrm{ord}} \log N$.

Corollary 6.11 (Complete-zero logarithmic confinement, effective form)**.** *With C_Z and N_0 as in Lemma 6.9, for every $N \ge N_0$,*

$$|\mathcal{C}_N| \le C_Z \log N, \tag{6.42}$$

and

$$\mathcal{M}_0(N) \le 4C_Z^2 (\log N)^2. \tag{6.43}$$

The factor 4 *comes only from the triangle inequality $|Z_n - \mathcal{C}_N| \le |Z_n| + |\mathcal{C}_N|$, giving $|Z_n - \mathcal{C}_N| \le 2C_Z \log N$ before squaring.*

In particular,

$$\mathcal{C}_N = O(\log N). \tag{6.44}$$

$$\mathcal{M}_0(N) = O\Big((\log N)^2\Big). \tag{6.45}$$

Proof. Lemma 6.9 gives $|Z_n| \le C_Z \log n$ for all $n \ge N_0$. Increasing N_0, if necessary, to absorb the finitely many values $n < N_0$, we have, for every $N \ge N_0$ and every $1 \le n \le N$,

$$|Z_n| \le C_Z \log N. \tag{6.46}$$

By definition,

$$|\mathcal{C}_N| \le \frac{1}{N} \sum_{n=1}^{N} |Z_n| \le C_Z \log N. \tag{6.47}$$

This proves (6.42) and hence (6.44). Thus

$$|\mathcal{C}_N| \le C_Z \log N. \tag{6.48}$$

Together with (6.46), this gives

$$|Z_n - \mathcal{C}_N| \le |Z_n| + |\mathcal{C}_N| \le 2C_Z \log N, \qquad 1 \le n \le N. \tag{6.49}$$

Squaring and averaging gives (6.43), and hence (6.45). □

6.1. **Specialization under the Riemann Hypothesis.**

Corollary 6.12 (Specialization under RH)**.** *Assume RH. Then*

$$\rho_k = \frac{1}{2} + i\gamma_k, \tag{6.50}$$

and hence

$$e^{\rho_k - \frac{1}{2}} = e^{i\gamma_k}. \tag{6.51}$$

Therefore

$$Z_N = S_N, \qquad \mathcal{C}_N = C_N, \qquad \mathcal{M}_0(N) = M_0(N). \tag{6.52}$$

Consequently,

$$M_0(N) = O\Big((\log N)^2\Big). \tag{6.53}$$

Proof. The identities follow directly from $e^{\rho_k - \frac{1}{2}} = e^{i\gamma_k}$ under RH and from the definitions of the complete-zero and unit-phase centroids and centered moments. The bound for $M_0(N)$ follows from Corollary 6.11. □

Here

$$M_0(N) = \frac{1}{N}\sum_{n=1}^{N} |S_n - C_N|^2. \tag{6.54}$$

6.2. **Roadmap of the logarithmic bound.** The logic of this section is as follows. Landau's fixed-real formula is applied directly at the single non-resonant value $X = e$, giving the height-truncated estimate for $\mathcal{Z}(T)$. The height-to-ordinal Lemma 6.9 then passes from T-truncation to N-truncation by isolating the boundary block B_N and bounding its multiplicity through Riemann–von Mangoldt. This gives $Z_N = O(\log N)$ with constants coming only from Landau at $X = e$ and from the unit-interval RVM bound. Averaging the prefix sums yields the complete-zero centroid and centered moment estimates. Under RH these complete-zero estimates specialize to the unit-phase objects S_N, C_N, and $M_0(N)$. The remaining subsections show that the centered moment is exactly a Brownian-bridge/Green-kernel weighted pair statistic.

6.3. **Pair-correlation form.** The preceding estimate has an exact weighted pair-correlation form. Since

$$Z_n = \sum_{k=1}^{n} e^{\rho_k - \frac{1}{2}}, \tag{6.55}$$

we have

$$|Z_n|^2 = \sum_{j,k \le n} e^{\rho_j - \frac{1}{2}} e^{\overline{\rho_k} - \frac{1}{2}}. \tag{6.56}$$

Moreover,

$$\mathcal{M}_0(N) = \frac{1}{N}\sum_{n=1}^{N} |Z_n|^2 - |\mathcal{C}_N|^2. \tag{6.57}$$

Using the triangular form of $\mathcal{C}_N$,

$$\mathcal{C}_N = \frac{1}{N}\sum_{k=1}^{N} (N - k + 1) e^{\rho_k - \frac{1}{2}}, \tag{6.58}$$

we obtain

$$\mathcal{M}_0(N) = \sum_{j,k=1}^{N} W_N(j,k) e^{\rho_j - \frac{1}{2}} e^{\overline{\rho_k} - \frac{1}{2}}, \tag{6.59}$$

where

$$W_N(j,k) = \frac{N - \max(j,k) + 1}{N} - \frac{(N-j+1)(N-k+1)}{N^2}. \tag{6.60}$$

Equivalently, setting

$$a = N - j + 1, \qquad b = N - k + 1, \tag{6.61}$$

we may write

$$W_N(j,k) = \frac{\min(a,b)}{N} - \frac{ab}{N^2}. \tag{6.62}$$

The trace of the centering kernel is explicit:

$$\sum_{j=1}^{N} W_N(j,j) = \frac{N^2-1}{6N}. \tag{6.63}$$

Combining (6.59) with Corollary 6.11, we obtain

$$\sum_{j,k=1}^{N} W_N(j,k) e^{\rho_j - \frac{1}{2}} e^{\overline{\rho_k} - \frac{1}{2}} = O\Big((\log N)^2\Big). \tag{6.64}$$

This is a Brownian-bridge weighted pair-correlation cancellation for the complete-zero walk.

Here and below, the phrase "Brownian-bridge" refers to the exact algebraic equality of the centering kernel with the finite Brownian-bridge covariance kernel. No probabilistic approximation, no Gaussian process limit, and no distributional convergence statement is being asserted at this point.

Under RH, the pair-correlation identity (6.59) becomes

$$M_0(N) = \sum_{j,k=1}^{N} W_N(j,k) e^{i(\gamma_j - \gamma_k)}. \tag{6.65}$$

In particular,

$$\sum_{j,k=1}^{N} W_N(j,k) e^{i(\gamma_j - \gamma_k)} = O\Big((\log N)^2\Big). \tag{6.66}$$

Lemma 6.13 (Diagonal and off-diagonal balance)**.** *Assume RH, so that the phase increments have unit modulus. Then*

$$M_0(N) = D_N + O_N, \tag{6.67}$$

where

$$D_N = \sum_{j=1}^{N} W_N(j,j) = \frac{N^2-1}{6N}, \tag{6.68}$$

and

$$O_N = \sum_{\substack{1\le j,k\le N\\ j\neq k}} W_N(j,k) e^{i(\gamma_j-\gamma_k)}. \tag{6.69}$$

Consequently, if $M_0(N) = O((\log N)^2)$*, then*

$$O_N = -\frac{N^2-1}{6N} + O\Big((\log N)^2\Big). \tag{6.70}$$

Proof. Under RH,

$$|e^{i\gamma_j}| = 1, \tag{6.71}$$

and hence the diagonal part of the pair statistic is

$$D_N = \sum_{j=1}^{N} W_N(j,j). \tag{6.72}$$

From

$$W_N(j,k) = \frac{N-\max(j,k)+1}{N} - \frac{(N-j+1)(N-k+1)}{N^2}, \tag{6.73}$$

we get

$$W_N(j,j) = \frac{N-j+1}{N} - \frac{(N-j+1)^2}{N^2}. \tag{6.74}$$

Put

$$a = N-j+1. \tag{6.75}$$

As j runs from 1 to N, a runs from N down to 1. Therefore

$$D_N = \sum_{a=1}^{N} \left(\frac{a}{N} - \frac{a^2}{N^2}\right). \tag{6.76}$$

Using

$$\sum_{a=1}^{N} a = \frac{N(N+1)}{2}, \qquad \sum_{a=1}^{N} a^2 = \frac{N(N+1)(2N+1)}{6}, \tag{6.77}$$

we obtain

$$D_N = \frac{N+1}{2} - \frac{(N+1)(2N+1)}{6N} = \frac{N^2-1}{6N}. \tag{6.78}$$

Since $M_0(N) = D_N + O_N$, the claimed off-diagonal identity follows immediately from $M_0(N) = O((\log N)^2)$. □

Thus the logarithmic bound on the centered moment is much stronger than the diagonal size. The diagonal contribution has order N, while the full centered moment is only $O((\log N)^2)$. Therefore the off-diagonal contribution must cancel the diagonal to accuracy $O((\log N)^2)$ for this specific Brownian-bridge kernel.

In the complete-zero setting the diagonal contains the factor

$$e^{\rho_j-\frac{1}{2}}e^{\overline{\rho_j}-\frac{1}{2}} = e^{2\beta_j-1}. \tag{6.79}$$

Thus the complete-zero diagonal is

$$\sum_{j=1}^{N} W_N(j,j)e^{2\beta_j-1}. \tag{6.80}$$

Only under RH does this reduce to

$$\sum_{j=1}^{N} W_N(j,j) = \frac{N^2-1}{6N}. \tag{6.81}$$

Thus the RH specialization recovers the original centered phase walk and identifies its logarithmic quadratic confinement with Brownian-bridge weighted off-diagonal cancellation.

6.4. **The Brownian-bridge Green kernel forced by centering.** The logarithmic bound follows from Landau's fixed-real formula at $X = e$, together with the elementary passage from endpoint bounds to centered quadratic bounds.

The estimate itself is classical in origin. The structural point of the present formulation is different: the centered phase-walk moment forces an exact Green-kernel pair statistic. Thus the centroid geometry converts a one-dimensional phase walk into a weighted pair-correlation statistic with Brownian-bridge covariance kernel.

Starting from the complete-zero phase-walk centroid $\mathcal{C}_N$, the centered quadratic moment $\mathcal{M}_0(N)$ produces an exact pair-correlation identity with the explicit weight (6.60). After the change of variables (6.61), this weight becomes (6.62). This is the standard discrete Brownian-bridge covariance kernel, equivalently the Green kernel of the discrete Dirichlet Laplacian on the corresponding finite interval.

Thus the structural point is the way the centered geometry of the zeta-zero phase walk forces exactly this Green/Brownian-bridge kernel and therefore converts the centroid problem into a precise weighted pair-correlation statistic of the zeta zeros.

This is an algebraic identity between finite quadratic forms and kernels. It is not a claim that the zeta-zero phase walk converges to a Brownian bridge, nor that the modal coefficients are Gaussian. Any probabilistic or distributional strengthening would require additional hypotheses and error estimates not used in the present argument.

In short:

$$\begin{aligned} &\text{centered phase-walk geometry} \Longrightarrow \text{Brownian-bridge Green kernel} \\ &\qquad\qquad\qquad\qquad\qquad\Longrightarrow \text{weighted pair correlation of zeta zeros.} \end{aligned} \tag{6.82}$$

Landau's fixed-real formula at $X = e$ supplies the logarithmic confinement baseline. The new observation is that the centered phase-walk moment realizes this confinement through an exact Brownian-bridge/Green-kernel pair statistic, and that the same Green-modal language gives a natural expression of the Montgomery–GUE form factor.

In the RH/unit-phase specialization, the diagonal part of the Brownian-bridge pair statistic has size (6.63). The logarithmic bound therefore expresses a near-total off-diagonal cancellation against

this specific Green kernel. The structural contribution is the identification of the precise kernel and geometric object through which Landau non-resonance appears in the phase-walk formulation.

6.5. **Spectral lifting and the Green-kernel modal decomposition.** The centered phase-walk moment is not only a scalar statistic. It is an exact Hermitian Green-kernel energy, and the Green kernel admits a complete spectral decomposition.

Lemma 6.14 (Lifted phase representation)**.** *For every ordinate $\gamma_k > 0$, define*

$$B_k = \left\lfloor \frac{\gamma_k}{2\pi} \right\rfloor, \qquad \theta_k = \gamma_k - 2\pi B_k \in [0, 2\pi). \tag{6.83}$$

Then

$$\gamma_k = 2\pi B_k + \theta_k, \tag{6.84}$$

and

$$e^{i\gamma_k} = e^{i\theta_k}. \tag{6.85}$$

Moreover, for every pair j, k,

$$\gamma_j - \gamma_k = 2\pi(B_j - B_k) + (\theta_j - \theta_k). \tag{6.86}$$

Consequently, for every real t,

$$e^{it(\gamma_j - \gamma_k)} = e^{i2\pi t(B_j - B_k)} e^{it(\theta_j - \theta_k)}. \tag{6.87}$$

Proof. The identity $\gamma_k = 2\pi B_k + \theta_k$ is the Euclidean decomposition of the real number γ_k modulo 2π. Since $B_k \in \mathbb{Z}$,

$$e^{i\gamma_k} = e^{i(2\pi B_k + \theta_k)} = e^{i2\pi B_k} e^{i\theta_k} = e^{i\theta_k}. \tag{6.88}$$

Subtracting the decompositions for γ_j and γ_k gives (6.86). Multiplying by it and exponentiating gives (6.87). □

Thus the ordinary phase walk

$$S_N = \sum_{k=1}^{N} e^{i\gamma_k} \tag{6.89}$$

can be formed from the angular representatives alone:

$$S_N = \sum_{k=1}^{N} e^{i\theta_k}. \tag{6.90}$$

However, the angular representatives alone do not determine γ_k. The additional winding integers B_k restore the full ordinate:

$$\gamma_k = 2\pi B_k + \theta_k. \tag{6.91}$$

For $t = 1$, the winding is invisible:

$$e^{i\gamma_k} = e^{i\theta_k}. \tag{6.92}$$

For general real t,

$$e^{it\gamma_k} = e^{i2\pi tB_k} e^{it\theta_k}, \tag{6.93}$$

so the winding data are needed to reconstruct the true real-frequency phase. In finite-N verification, B_k is not an additional hypothesis: it is computed directly from the ordinate by (6.83). For Montgomery scaling $t = \tau \log(T_N/2\pi) + y$, retaining B_k is essential because the factor $e^{i2\pi tB_k}$ is generally nontrivial.

There are two reversed-index conventions in the paper. For pair variables j, k, we use

$$r = N - j + 1, \qquad s = N - k + 1. \tag{6.94}$$

In these variables the Green/Brownian weight is

$$K_N(r,s) = \frac{\min(r,s)}{N} - \frac{rs}{N^2}, \qquad 1 \le r, s \le N-1. \tag{6.95}$$

Thus $W_N(j,k) = K_N(r,s)$. For a single endpoint vector indexed by k, we use

$$r = N - k + 1, \qquad x_r(t) = e^{it\gamma_{N-r+1}}. \tag{6.96}$$

These are the same reversal convention applied respectively to a pair of indices and to a single index.

This is, as a finite matrix identity, the covariance kernel of the discrete Brownian bridge, equivalently the Green kernel of the discrete Dirichlet Laplacian on the finite interval. The terminology identifies the kernel; it does not introduce a random process into the proof.

Lemma 6.15 (Orthonormality of the Dirichlet sine basis)**.** *For $1 \le m, r \le N-1$, define*

$$\phi_m(r) = \sqrt{\frac{2}{N}} \sin\left(\frac{\pi mr}{N}\right). \tag{6.97}$$

Then

$$\sum_{r=1}^{N-1} \phi_m(r)\overline{\phi_\ell(r)} = \delta_{m\ell}. \tag{6.98}$$

Since the functions ϕ_m are real-valued, this is equivalently

$$\sum_{r=1}^{N-1} \phi_m(r)\phi_\ell(r) = \delta_{m\ell}.$$

Consequently, $\{\phi_m\}_{m=1}^{N-1}$ is an orthonormal basis of $\mathbb{C}^{N-1}$.

Proof. It is enough to prove

$$\sum_{r=1}^{N-1} \sin\left(\frac{\pi mr}{N}\right) \sin\left(\frac{\pi \ell r}{N}\right) = \frac{N}{2}\delta_{m\ell}. \tag{6.99}$$

Using

$$\sin a \sin b = \frac{1}{2}(\cos(a-b) - \cos(a+b)),$$

we obtain

$$\sum_{r=1}^{N-1} \sin\left(\frac{\pi mr}{N}\right) \sin\left(\frac{\pi \ell r}{N}\right) = \frac{1}{2}\left(C_{m-\ell} - C_{m+\ell}\right), \tag{6.100}$$

where

$$C_q = \sum_{r=1}^{N-1} \cos\left(\frac{\pi q r}{N}\right). \tag{6.101}$$

We compute C_q explicitly. Let

$$\omega_q = e^{i\pi q/N}.$$

If $\omega_q \neq 1$, then

$$\sum_{r=1}^{N-1} e^{i\pi q r/N} = \sum_{r=1}^{N-1} \omega_q^r = \frac{\omega_q(1-\omega_q^{N-1})}{1-\omega_q}. \tag{6.102}$$

Taking real parts gives C_q. Equivalently, for $0 \le q \le 2N-1$,

$$C_q = \begin{cases} N-1, & q = 0, \\ 0, & q \text{ odd}, \\ -1, & q \text{ even},\ q \neq 0,\ 2N. \end{cases} \tag{6.103}$$

Also $C_{-q} = C_q$, since cosine is even.

In the present application,

$$1 \le m, \ell \le N-1.$$

Hence

$$0 \le |m-\ell| \le N-2, \qquad 2 \le m+\ell \le 2N-2. \tag{6.104}$$

Thus the exceptional value $q = 2N$ never occurs in $m+\ell$, and $q = 0$ occurs only in $m-\ell$ when $m = \ell$.

If $m = \ell$, then $m-\ell = 0$ and $m+\ell = 2m$, with

$$2 \le 2m \le 2N-2.$$

Therefore

$$C_{m-\ell} = C_0 = N-1, \qquad C_{m+\ell} = C_{2m} = -1.$$

Consequently,

$$\sum_{r=1}^{N-1} \sin^2\left(\frac{\pi m r}{N}\right) = \frac{1}{2}((N-1)-(-1)) = \frac{N}{2}. \tag{6.105}$$

If $m \neq \ell$, there are two cases. If m and ℓ have different parity, then $m-\ell$ and $m+\ell$ are odd, so

$$C_{m-\ell} = C_{m+\ell} = 0.$$

If m and ℓ have the same parity, then $m-\ell$ and $m+\ell$ are even. Since $m \neq \ell$, the first is nonzero, and the second is always nonzero. Both lie in the permitted range away from $2N$, so

$$C_{m-\ell} = C_{m+\ell} = -1.$$

In both subcases,

$$C_{m-\ell} - C_{m+\ell} = 0.$$

Hence

$$\sum_{r=1}^{N-1} \sin\left(\frac{\pi m r}{N}\right) \sin\left(\frac{\pi \ell r}{N}\right) = 0 \qquad (m \neq \ell).$$

Multiplying by the normalization factor $2/N$ gives

$$\sum_{r=1}^{N-1} \phi_m(r)\overline{\phi_\ell(r)} = \delta_{m\ell},$$

which is (6.98). Since there are $N-1$ orthonormal vectors in the $N-1$-dimensional space $\mathbb{C}^{N-1}$, they form an orthonormal basis.

□

The Green kernel has the spectral decomposition

$$K_N(r,s) = \sum_{m=1}^{N-1} \lambda_m \phi_m(r)\phi_m(s), \tag{6.106}$$

where

$$\lambda_m = \frac{1}{4N \sin^2\left(\frac{\pi m}{2N}\right)}. \tag{6.107}$$

Proposition 6.16 (Elementary Green-kernel limits)**.** *As $N \to \infty$,*

$$\frac{N^2-1}{6N} \sim \frac{N}{6}. \tag{6.108}$$

For each fixed $m \geq 1$,

$$\lambda_m = \frac{1}{4N \sin^2\left(\frac{\pi m}{2N}\right)} \sim \frac{N}{\pi^2 m^2}. \tag{6.109}$$

Proof. The first statement is immediate from $(N^2-1)/(6N) = N/6 - 1/(6N)$. For fixed m, $\sin(\pi m/(2N)) \sim \pi m/(2N)$, which gives (6.109). □

Under RH, the phase increments are

$$q_k = e^{i\gamma_k}. \tag{6.110}$$

Define the modal coefficients

$$A_m(N) = \sum_{r=1}^{N-1} \phi_m(r) q_{N-r+1}. \tag{6.111}$$

Equivalently,

$$A_m(N) = \sum_{r=1}^{N-1} \sqrt{\frac{2}{N}} \sin\left(\frac{\pi m r}{N}\right) e^{i\gamma_{N-r+1}}. \tag{6.112}$$

Then the centered quadratic moment has the exact modal identity

$$M_0(N) = \sum_{m=1}^{N-1} \lambda_m |A_m(N)|^2. \tag{6.113}$$

For the complete-zero walk, put

$$q_k = e^{\rho_k - \frac{1}{2}}. \tag{6.114}$$

Then

$$\mathcal{M}_0(N) = \sum_{m=1}^{N-1} \lambda_m |\mathcal{A}_m(N)|^2, \tag{6.115}$$

where

$$\mathcal{A}_m(N) = \sum_{r=1}^{N-1} \phi_m(r) e^{\rho_{N-r+1} - \frac{1}{2}}. \tag{6.116}$$

Expanding the squared modulus gives

$$|\mathcal{A}_m(N)|^2 = \sum_{r,s=1}^{N-1} \phi_m(r)\phi_m(s) e^{\rho_{N-r+1} - \frac{1}{2}} e^{\overline{\rho_{N-s+1}} - \frac{1}{2}}. \tag{6.117}$$

Under RH this reduces to the unit phase identity

$$e^{\rho_{N-r+1} - \frac{1}{2}} e^{\overline{\rho_{N-s+1}} - \frac{1}{2}} = e^{i\gamma_{N-r+1}} e^{-i\gamma_{N-s+1}} = e^{i(\gamma_{N-r+1} - \gamma_{N-s+1})}. \tag{6.118}$$

Lemma 6.17 (Complete-zero modal Parseval bound)**.**

$$\sum_{m=1}^{N-1} |\mathcal{A}_m(N)|^2 = \sum_{k=2}^{N} e^{2\beta_k - 1}. \tag{6.119}$$

$$\sum_{m=1}^{N-1} |\mathcal{A}_m(N)|^2 \le e(N-1), \tag{6.120}$$

and hence

$$|\mathcal{A}_m(N)| \le \sqrt{e(N-1)}. \tag{6.121}$$

Under RH this becomes

$$\sum_{m=1}^{N-1} |\mathcal{A}_m(N)|^2 = N - 1, \qquad |\mathcal{A}_m(N)| \le \sqrt{N-1}. \tag{6.122}$$

Proof. Apply Parseval to the vector

$$z_r = e^{\rho_{N-r+1} - \frac{1}{2}}. \tag{6.123}$$

Then

$$\sum_{m=1}^{N-1} |\mathcal{A}_m(N)|^2 = \sum_{r=1}^{N-1} |z_r|^2. \tag{6.124}$$

Since

$$\left| e^{\rho_k - \frac{1}{2}} \right|^2 = e^{2\beta_k - 1}, \tag{6.125}$$

we get

$$\sum_{m=1}^{N-1} |\mathcal{A}_m(N)|^2 = \sum_{k=2}^{N} e^{2\beta_k - 1}. \tag{6.126}$$

The bound follows from $0 < \beta_k < 1$. Under RH, $\beta_k = 1/2$, so every factor is 1. □

This shows that the Brownian-bridge identity is not merely a scalar rewriting of $M_0(N)$. It supplies a canonical Hermitian spectral frame for the centered phase-walk geometry. The scalar moment $M_0(N)$ is the total Green energy, while the quantities $|A_m(N)|^2$ are its modal energy components.

The lifted representation (6.83) ensures that the use of the circular phases $e^{i\gamma_k}$ does not destroy the real ordinate information, provided the winding integers B_k are retained. Therefore the Green-kernel modal representation can be formulated together with real pair differences (6.86).

6.6. **Green-generated Fourier pair statistics.** The preceding phase walk corresponds to the single frequency $t = 1$. For pair-correlation analysis one may allow the real Fourier parameter t. The corresponding Green-weighted statistic is

$$M_0(N;t) = \sum_{j,k=1}^{N} W_N(j,k) e^{it(\gamma_j - \gamma_k)}. \tag{6.127}$$

For $t = 1$, this recovers the original centered phase-walk statistic. With

$$A_m(N;t) = \sum_{r=1}^{N-1} \phi_m(r) e^{it\gamma_{N-r+1}}. \tag{6.128}$$

Lemma 6.18 (Uniform finite-N modal bounds)**.** *For every $N \geq 2$, every $1 \leq m \leq N-1$, and every real t,*

$$|A_m(N;t)| \leq \sqrt{N-1}. \tag{6.129}$$

Moreover,

$$\sum_{m=1}^{N-1} |A_m(N;t)|^2 = N - 1. \tag{6.130}$$

Consequently,

$$\frac{1}{N} \sum_{m=1}^{N-1} |A_m(N;t)|^2 \leq 1. \tag{6.131}$$

Proof. By Cauchy–Schwarz,

$$|A_m(N;t)| \leq \left(\sum_{r=1}^{N-1} |\phi_m(r)|^2 \right)^{1/2} \left(\sum_{r=1}^{N-1} |e^{it\gamma_{N-r+1}}|^2 \right)^{1/2}. \tag{6.132}$$

The sine functions ϕ_m form an orthonormal basis of $\mathbb{C}^{N-1}$, hence

$$\sum_{r=1}^{N-1} |\phi_m(r)|^2 = 1. \tag{6.133}$$

Also,

$$|e^{it\gamma_{N-r+1}}| = 1, \tag{6.134}$$

so

$$\sum_{r=1}^{N-1} |e^{it\gamma_{N-r+1}}|^2 = N-1. \tag{6.135}$$

This gives (6.129).
For the second identity, set

$$x_r(t) = e^{it\gamma_{N-r+1}}. \tag{6.136}$$

The coefficients $A_m(N;t)$ are precisely the sine-basis coefficients of the vector $x(t)$. By Parseval,

$$\sum_{m=1}^{N-1} |A_m(N;t)|^2 = \sum_{r=1}^{N-1} |x_r(t)|^2 = N-1. \tag{6.137}$$

□

Since the estimate is pointwise in t, it applies uniformly to the Montgomery scaling

$$t = \tau \log(T_N/2\pi) + y. \tag{6.138}$$

Consequently,

$$\int_{-\infty}^{\infty} e^{-2|y|} |A_m(N; \tau \log(T_N/2\pi) + y)|^2 \, dy \leq N-1, \tag{6.139}$$

because

$$\int_{-\infty}^{\infty} e^{-2|y|} \, dy = 1. \tag{6.140}$$

The exact modal identity is

$$M_0(N;t) = \sum_{m=1}^{N-1} \lambda_m |A_m(N;t)|^2. \tag{6.141}$$

This is the t-deformed Green energy identity. In lifted variables,

$$e^{it(\gamma_j - \gamma_k)} = e^{it[2\pi(B_j - B_k) + (\theta_j - \theta_k)]}. \tag{6.142}$$

Thus the t-deformed Green statistic is a Fourier statistic of the lifted real pair differences, not merely of circular phases. It is, however, Green-weighted. The unweighted Montgomery form factor is obtained from the endpoint projection identity below, rather than from a single scalar value of $M_0(N;t)$.

6.7. **Endpoint projection and Green deweighting.** For later Montgomery scaling, retain

$$T_N = \gamma_N, \qquad L_N = \frac{\log(T_N/2\pi)}{2\pi}. \tag{6.143}$$

For every real t, define

$$q_k(t) = e^{it\gamma_k}. \tag{6.144}$$

In reversed variables,

$$r = N - k + 1, \tag{6.145}$$

put

$$x_r(t) = q_{N-r+1}(t) = e^{it\gamma_{N-r+1}}, \qquad 1 \le r \le N-1. \tag{6.146}$$

The Green/Brownian sine basis is

$$\phi_m(r) = \sqrt{\frac{2}{N}} \sin\left(\frac{\pi m r}{N}\right), \qquad 1 \le m, r \le N-1. \tag{6.147}$$

The modal space is $\mathbb{C}^{N-1}$: the endpoint projection below separates the first term $e^{it\gamma_1}$, and the modal vector is built from $x_r(t) = e^{it\gamma_{N-r+1}}$, $1 \le r \le N-1$, corresponding to the ordinates $k = 2, \dots, N$.

The modal coefficients are

$$A_m(N;t) = \sum_{r=1}^{N-1} \phi_m(r) x_r(t) = \sum_{r=1}^{N-1} \phi_m(r) e^{it\gamma_{N-r+1}}. \tag{6.148}$$

Let

$$A(N;t) = (A_1(N;t), \dots, A_{N-1}(N;t))^T. \tag{6.149}$$

The Green-modal energy is

$$\mathcal{E}_N(t) = A(N;t)^* \Lambda_N A(N;t) = \sum_{m=1}^{N-1} \lambda_m |A_m(N;t)|^2, \tag{6.150}$$

where

$$\Lambda_N = \operatorname{diag}(\lambda_1, \dots, \lambda_{N-1}) \tag{6.151}$$

and

$$\lambda_m = \frac{1}{4N \sin^2\left(\frac{\pi m}{2N}\right)}. \tag{6.152}$$

In the notation of the centered t-phase walk,

$$\mathcal{E}_N(t) = M_0(N;t). \tag{6.153}$$

Lemma 6.19 (Exact modal inversion)**.** *Let*

$$x_r(t) = e^{it\gamma_{N-r+1}}, \qquad 1 \le r \le N-1, \tag{6.154}$$

and define

$$A_m(N;t) = \sum_{r=1}^{N-1} \phi_m(r) x_r(t). \tag{6.155}$$

Then

$$x_r(t) = \sum_{m=1}^{N-1} A_m(N;t)\phi_m(r), \qquad 1 \le r \le N-1. \tag{6.156}$$

Proof. The result is the expansion of $x(t) \in \mathbb{C}^{N-1}$ in the orthonormal basis $\{\phi_m\}_{m=1}^{N-1}$. Since the ϕ_m are real-valued,

$$A_m(N;t) = \sum_{r=1}^{N-1} \phi_m(r) x_r(t) \tag{6.157}$$

are precisely the basis coefficients. Hence (6.156) follows. □

Lemma 6.20 (General modal deweighting)**.** *Let $a = (a_1, \dots, a_{N-1})^T \in \mathbb{C}^{N-1}$, and define*

$$\eta_m = \sum_{r=1}^{N-1} \phi_m(r) a_r, \qquad 1 \le m \le N-1. \tag{6.158}$$

Then, for every real t,

$$a^* x(t) = \eta^* A(N;t). \tag{6.159}$$

Equivalently,

$$a^* x(t) = (\Lambda_N^{-1}\eta)^* \Lambda_N A(N;t). \tag{6.160}$$

Consequently,

$$|a^* x(t)|^2 \le (\eta^* \Lambda_N^{-1} \eta)\, \mathcal{E}_N(t). \tag{6.161}$$

Proof. By Lemma 6.15, both $x(t)$ and a have exact sine-basis expansions,

$$x(t) = \sum_{m=1}^{N-1} A_m(N;t)\phi_m, \qquad a = \sum_{m=1}^{N-1} \eta_m \phi_m. \tag{6.162}$$

Taking the Hermitian inner product gives (6.159). Since

$$\eta^* A(N;t) = (\Lambda_N^{-1}\eta)^* \Lambda_N A(N;t), \tag{6.163}$$

(6.160) follows. Applying Cauchy–Schwarz in the Λ_N-metric gives

$$|\eta^* A(N;t)|^2 \le (\eta^* \Lambda_N^{-1} \eta)(A(N;t)^* \Lambda_N A(N;t)), \tag{6.164}$$

which is (6.161). □

Lemma 6.21 (Constant-mode projection)**.** *Define*

$$u_m = \sum_{r=1}^{N-1} \phi_m(r). \tag{6.165}$$

Then

$$u_m = \begin{cases} \sqrt{\dfrac{2}{N}} \cot\left(\dfrac{\pi m}{2N}\right), & m \text{ odd}, \\ 0, & m \text{ even}. \end{cases} \tag{6.166}$$

Proof. Using

$$\sum_{r=1}^{N-1} \sin(rx) = \frac{\sin\left(\frac{(N-1)x}{2}\right) \sin\left(\frac{Nx}{2}\right)}{\sin\left(\frac{x}{2}\right)}, \tag{6.167}$$

with

$$x = \frac{\pi m}{N}, \tag{6.168}$$

we obtain

$$\sum_{r=1}^{N-1} \sin\left(\frac{\pi m r}{N}\right) = \frac{\sin\left(\frac{\pi m}{2} - \frac{\pi m}{2N}\right) \sin\left(\frac{\pi m}{2}\right)}{\sin\left(\frac{\pi m}{2N}\right)}. \tag{6.169}$$

If m is even, then

$$\sin\left(\frac{\pi m}{2}\right) = 0, \tag{6.170}$$

and the sum is zero. If m is odd, then

$$\sin\left(\frac{\pi m}{2}\right) = (-1)^{(m-1)/2}, \tag{6.171}$$

and

$$\sin\left(\frac{\pi m}{2} - \frac{\pi m}{2N}\right) = (-1)^{(m-1)/2} \cos\left(\frac{\pi m}{2N}\right). \tag{6.172}$$

Therefore

$$\sum_{r=1}^{N-1} \sin\left(\frac{\pi m r}{N}\right) = \frac{\cos\left(\frac{\pi m}{2N}\right)}{\sin\left(\frac{\pi m}{2N}\right)} = \cot\left(\frac{\pi m}{2N}\right). \tag{6.173}$$

Multiplying by the normalization factor $\sqrt{2/N}$ gives (6.166). □

Lemma 6.22 (Endpoint recovery from Green-modal coordinates)**.** *Let*

$$u_m = \sum_{r=1}^{N-1} \phi_m(r). \tag{6.174}$$

Then

$$\sum_{k=2}^{N} e^{it\gamma_k} = \sum_{m=1}^{N-1} u_m A_m(N;t). \tag{6.175}$$

Therefore

$$\sum_{k=1}^{N} e^{it\gamma_k} = e^{it\gamma_1} + \sum_{m=1}^{N-1} u_m A_m(N;t). \tag{6.176}$$

Since

$$u_m = \begin{cases} \sqrt{\dfrac{2}{N}} \cot\left(\dfrac{\pi m}{2N}\right), & m \text{ odd}, \\ 0, & m \text{ even}, \end{cases} \tag{6.177}$$

this may also be written as

$$\sum_{k=1}^{N} e^{it\gamma_k} = e^{it\gamma_1} + \sum_{\substack{1 \le m \le N-1 \\ m \text{ odd}}} u_m A_m(N;t). \tag{6.178}$$

Proof. By exact modal inversion,

$$x_r(t) = \sum_{m=1}^{N-1} A_m(N;t)\phi_m(r). \tag{6.179}$$

Summing over r gives

$$\sum_{r=1}^{N-1} x_r(t) = \sum_{m=1}^{N-1} A_m(N;t) \sum_{r=1}^{N-1} \phi_m(r) = \sum_{m=1}^{N-1} u_m A_m(N;t). \tag{6.180}$$

Since $x_r(t) = e^{it\gamma_{N-r+1}}$, the change of variables $k = N - r + 1$ gives

$$\sum_{r=1}^{N-1} x_r(t) = \sum_{k=2}^{N} e^{it\gamma_k}. \tag{6.181}$$

Adding the missing endpoint $e^{it\gamma_1}$ gives

$$\sum_{k=1}^{N} e^{it\gamma_k} = e^{it\gamma_1} + \sum_{m=1}^{N-1} u_m A_m(N;t). \tag{6.182}$$

The parity reduction follows from Lemma 6.21. □

The even modes $A_m(N;t)$ are not zero in general. They disappear from the endpoint sum only because their projection coefficient u_m vanishes. They still contribute to the centered Green energy

$$M_0(N;t) = \sum_{m=1}^{N-1} \lambda_m |A_m(N;t)|^2. \tag{6.183}$$

Thus

(6.184) the endpoint projection sees only odd Green modes,

whereas

(6.185) the centered Green energy sees all modes.

6.8. **Unconditional Green deweighting of the endpoint projection.**
Let $u = (u_1, \dots, u_{N-1})^T$.

Theorem 6.23 (Unconditional finite Green deweighting)**.** *For every $N \geq 2$ and every real t,*

$$\left| \sum_{k=2}^{N} e^{it\gamma_k} \right|^2 \leq 2N\, \mathcal{E}_N(t). \tag{6.186}$$

Equivalently,

$$\frac{1}{N} \left| \sum_{k=2}^{N} e^{it\gamma_k} \right|^2 \leq 2\, \mathcal{E}_N(t). \tag{6.187}$$

Proof. By the endpoint projection identity without the first endpoint,

$$\sum_{k=2}^{N} e^{it\gamma_k} = u^* A(N;t). \tag{6.188}$$

Since Λ_N is positive definite,

$$u^* A(N;t) = (\Lambda_N^{-1/2} u)^* (\Lambda_N^{1/2} A(N;t)). \tag{6.189}$$

By Cauchy–Schwarz,

$$|u^* A(N;t)|^2 \leq (u^* \Lambda_N^{-1} u)(A(N;t)^* \Lambda_N A(N;t)). \tag{6.190}$$

It remains to compute $u^* \Lambda_N^{-1} u$. From Lemma 6.21,

$$u^* \Lambda_N^{-1} u = \sum_{\substack{1 \leq m \leq N-1 \\ m \text{ odd}}} |u_m|^2 \lambda_m^{-1}. \tag{6.191}$$

Since

$$\lambda_m^{-1} = 4N \sin^2\left(\frac{\pi m}{2N}\right), \tag{6.192}$$

we get, for odd m,

$$|u_m|^2 \lambda_m^{-1} = \frac{2}{N} \cot^2\left(\frac{\pi m}{2N}\right) 4N \sin^2\left(\frac{\pi m}{2N}\right) = 8 \cos^2\left(\frac{\pi m}{2N}\right). \tag{6.193}$$

It remains to sum these terms over odd m. Write $\cos^2 x = (1 + \cos(2x))/2$. For odd m, $2x = \pi m/N$. We use the finite cosine sum

$$\sum_{\ell=1}^{J} \cos((2\ell - 1)x) = \frac{\sin(2Jx)}{2 \sin x}. \tag{6.194}$$

If $N = 2J$, then

$$\sum_{\substack{1 \le m \le N-1 \\ m \text{ odd}}} \cos\left(\frac{\pi m}{N}\right) = \sum_{\ell=1}^{J} \cos\left(\frac{(2\ell-1)\pi}{2J}\right) = \frac{\sin \pi}{2 \sin(\pi/(2J))} = 0. \tag{6.195}$$

Hence

$$\sum_{\substack{1 \le m \le N-1 \\ m \text{ odd}}} \cos^2\left(\frac{\pi m}{2N}\right) = \frac{1}{2} J = \frac{N}{4}. \tag{6.196}$$

If $N = 2J + 1$, then

$$\sum_{\substack{1 \le m \le N-1 \\ m \text{ odd}}} \cos\left(\frac{\pi m}{N}\right) = \sum_{\ell=1}^{J} \cos\left(\frac{(2\ell-1)\pi}{2J+1}\right) = \frac{\sin(2J\pi/(2J+1))}{2 \sin(\pi/(2J+1))} = \frac{1}{2}. \tag{6.197}$$

Therefore

$$\sum_{\substack{1 \le m \le N-1 \\ m \text{ odd}}} \cos^2\left(\frac{\pi m}{2N}\right) = \frac{1}{2} J + \frac{1}{4} = \frac{2J+1}{4} = \frac{N}{4}. \tag{6.198}$$

Thus, for every $N \ge 2$,

$$\sum_{\substack{1 \le m \le N-1 \\ m \text{ odd}}} \cos^2\left(\frac{\pi m}{2N}\right) = \frac{N}{4}, \tag{6.199}$$

and we obtain

$$u^* \Lambda_N^{-1} u = 2N. \tag{6.200}$$

Hence

$$\left| \sum_{k=2}^{N} e^{it\gamma_k} \right|^2 = |u^* A(N;t)|^2 \le 2N \, A(N;t)^* \Lambda_N A(N;t) = 2N \, \mathcal{E}_N(t). \tag{6.201}$$

□

This theorem is unconditional: it uses only the ordinates γ_k, the finite sine basis, and the Green metric. As a growth estimate it is generally weaker than the logarithmic endpoint control obtained from Landau's fixed-real formula at $X = e$; its role is to record the exact Green deweighting of the endpoint functional.

The coefficient vector $A(N;t)$ determines the endpoint projection exactly through $u^*A(N;t)$. By contrast, the scalar Green energy

$$\mathcal{E}_N(t) = A(N;t)^*\Lambda_N A(N;t) \tag{6.202}$$

does not determine the endpoint projection by itself, since it discards modal direction and phase information. It gives a one-sided estimate, not an exact inversion. The exact Green-metric recovery identity is

$$u^*A(N;t) = (\Lambda_N^{-1}u)^*\Lambda_N A(N;t). \tag{6.203}$$

Thus the scalar Green energy controls the endpoint projection from above. Exact endpoint recovery uses the coefficient vector $A(N;t)$, equivalently the weighted modal data $\Lambda_N A(N;t)$, through the linear functional in (6.203).

Corollary 6.24 (Centroid deweighting)**.** *For real t, define the t-deformed triangular centroid*

$$C_N(t) = \frac{1}{N}\sum_{k=1}^{N}(N-k+1)e^{it\gamma_k}. \tag{6.204}$$

Let $v = (v_1, \dots, v_{N-1})^T$ be given in reversed variables by

$$v_r = \frac{r}{N}, \qquad 1 \le r \le N-1. \tag{6.205}$$

and define its sine coefficients by

$$w_m = \sum_{r=1}^{N-1} v_r \phi_m(r). \tag{6.206}$$

Then

$$C_N(t) - e^{it\gamma_1} = w^*A(N;t). \tag{6.207}$$

Moreover,

$$w^*\Lambda_N^{-1}w = N-1, \tag{6.208}$$

and therefore

$$|C_N(t) - e^{it\gamma_1}|^2 \le (N-1)\mathcal{E}_N(t). \tag{6.209}$$

Proof. Using $r = N-k+1$ for $k = 2, \dots, N$, the definition (6.204) gives

$$C_N(t) - e^{it\gamma_1} = \sum_{r=1}^{N-1}\frac{r}{N}x_r(t) = v^*x(t). \tag{6.210}$$

By exact modal inversion, $x(t) = \sum_m A_m(N;t)\phi_m$, while $v = \sum_m w_m\phi_m$. Hence $v^*x(t) = w^*A(N;t)$, proving (6.207).

It remains to compute the Green dual norm. In physical variables, (6.215) gives $K_N^{-1} = NL_N$, where L_N is the finite Dirichlet Laplacian. Hence

$$w^*\Lambda_N^{-1}w = v^*K_N^{-1}v = N\,v^*L_Nv. \tag{6.211}$$

For $v_r = r/N$,

$$v^*L_Nv = v_1^2 + \sum_{r=1}^{N-2}(v_{r+1}-v_r)^2 + v_{N-1}^2 = \frac{N-1}{N}. \tag{6.212}$$

Thus $w^*\Lambda_N^{-1}w = N-1$. Applying Cauchy–Schwarz in the Λ_N-metric gives

$$|w^*A(N;t)|^2 \le (w^*\Lambda_N^{-1}w)(A(N;t)^*\Lambda_N A(N;t)) = (N-1)\mathcal{E}_N(t), \tag{6.213}$$

which is (6.209). □

This Green bound is generally weaker than the logarithmic centroid bound obtained by averaging Landau's endpoint estimate. Its role is structural: it gives the exact Green deweighting of the triangular centroid functional.

Remark 6.25 (Physical-space Green inverse). In reversed variables,

$$K_N(r,s) = \frac{\min(r,s)}{N} - \frac{rs}{N^2}, \qquad 1 \le r,s \le N-1. \tag{6.214}$$

This matrix is positive definite and satisfies

$$K_N^{-1} = NL_N, \tag{6.215}$$

where L_N is the discrete Dirichlet Laplacian

$$L_N = \begin{pmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \cdots & 0 \\ 0 & -1 & 2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & -1 \\ 0 & 0 & 0 & -1 & 2 \end{pmatrix}. \tag{6.216}$$

For the constant vector $\mathbf{1} = (1,\dots,1)^T$,

$$\mathbf{1}^*K_N^{-1}\mathbf{1} = N\,\mathbf{1}^*L_N\mathbf{1} = 2N. \tag{6.217}$$

Therefore, for every $x \in \mathbb{C}^{N-1}$,

$$|\mathbf{1}^*x|^2 \le (\mathbf{1}^*K_N^{-1}\mathbf{1})(x^*K_Nx) = 2N\,x^*K_Nx. \tag{6.218}$$

Taking $x_r = e^{it\gamma_{N-r+1}}$ gives

$$\left|\sum_{k=2}^{N} e^{it\gamma_k}\right|^2 \le 2N\,\mathcal{E}_N(t). \tag{6.219}$$

This is the operator inversion behind the deweighting estimate.

Let

$$t = \tau \log(T_N/2\pi) + y. \tag{6.220}$$

Define the endpoint form factor without the first endpoint by

$$F_N^{(2)}(\tau) = \frac{1}{N} \int_{-\infty}^{\infty} e^{-2|y|} \left| \sum_{k=2}^{N} e^{i(\tau \log(T_N/2\pi)+y)\gamma_k} \right|^2 dy. \tag{6.221}$$

Corollary 6.26 (Unconditional smoothed deweighting bound)**.** *For every $N \geq 2$ and every real τ,*

$$F_N^{(2)}(\tau) \leq 2 \int_{-\infty}^{\infty} e^{-2|y|} \mathcal{E}_N(\tau \log(T_N/2\pi) + y)\, dy. \tag{6.222}$$

Proof. Apply Theorem 6.23 pointwise with $t = \tau \log(T_N/2\pi) + y$. Then integrate against $e^{-2|y|}dy$ and divide by N. □

Define the full smoothed endpoint form factor by

$$F_N(\tau) = \frac{1}{N} \int_{-\infty}^{\infty} e^{-2|y|} \left| \sum_{k=1}^{N} e^{i(\tau \log(T_N/2\pi)+y)\gamma_k} \right|^2 dy. \tag{6.223}$$

Corollary 6.27 (Full endpoint bound)**.** *For every $N \geq 2$, every real τ, and every $\eta > 0$,*

$$F_N(\tau) \leq 2(1+\eta) \int_{-\infty}^{\infty} e^{-2|y|} \mathcal{E}_N(\tau \log(T_N/2\pi) + y)\, dy + \frac{1+\eta^{-1}}{N}. \tag{6.224}$$

Proof. Use $|a+b|^2 \leq (1+\eta)|a|^2 + (1+\eta^{-1})|b|^2$ with

$$t = \tau \log(T_N/2\pi) + y, \qquad a = \sum_{k=2}^{N} e^{it\gamma_k}, \qquad b = e^{it\gamma_1}. \tag{6.225}$$

Here $|b| = 1$, and the a-term is controlled by Corollary 6.26. □

Corollary 6.28 (Asymptotic deweighting consequence)**.** *For every fixed $\tau \in \mathbb{R}$,*

$$\limsup_{N\to\infty} F_N(\tau) \leq 2 \limsup_{N\to\infty} \int_{-\infty}^{\infty} e^{-2|y|} \mathcal{E}_N(\tau \log(T_N/2\pi) + y)\, dy, \tag{6.226}$$

with values allowed in $[0, \infty]$. More generally, if

$$\limsup_{N\to\infty} \int_{-\infty}^{\infty} e^{-2|y|} \mathcal{E}_N(\tau \log(T_N/2\pi) + y)\, dy \leq L(\tau), \tag{6.227}$$

then

$$\limsup_{N\to\infty} F_N(\tau) \leq 2L(\tau). \tag{6.228}$$

If the preceding bound holds uniformly for τ in a compact interval, then the same lim sup *estimate holds uniformly on that interval.*

Proof. From Corollary 6.27,

$$F_N(\tau) \le 2(1+\eta)\int_{-\infty}^{\infty} e^{-2|y|}\mathcal{E}_N(\tau\log(T_N/2\pi)+y)\,dy + \frac{1+\eta^{-1}}{N}. \tag{6.229}$$

Taking $\limsup_{N\to\infty}$ gives

$$\limsup_{N\to\infty} F_N(\tau) \le 2(1+\eta)\limsup_{N\to\infty}\int_{-\infty}^{\infty} e^{-2|y|}\mathcal{E}_N(\tau\log(T_N/2\pi)+y)\,dy. \tag{6.230}$$

Letting $\eta\downarrow 0$ gives the result. □

The deweighting estimate gives one-sided control of the endpoint projection by the Green energy. Exact recovery of the endpoint projection is vectorial:

$$u^*A(N;t) = (\Lambda_N^{-1}u)^*\Lambda_N A(N;t). \tag{6.231}$$

Thus the scalar energy

$$A(N;t)^*\Lambda_N A(N;t) \tag{6.232}$$

controls the size of the endpoint projection, while the weighted modal data

$$\Lambda_N A(N;t) \tag{6.233}$$

contain the phase information needed for exact recovery.

6.9. **The Green-modal form factor and the Montgomery–GUE target.** We now connect the Green/Brownian modal framework with the Montgomery pair-correlation form factor [13].

For the unsmoothed Montgomery scaling put

$$t = 2\pi\tau L_N = \tau\log(T_N/2\pi). \tag{6.234}$$

Then

$$e^{it\gamma_k} = e^{i\tau\log(T_N/2\pi)\gamma_k} = (T_N/2\pi)^{i\tau\gamma_k}. \tag{6.235}$$

Define the unsmoothed normalized form factor

$$\mathfrak{F}_N(\tau) = \frac{1}{N}\left|\sum_{k=1}^{N} e^{i2\pi\tau L_N\gamma_k}\right|^2. \tag{6.236}$$

Using (6.178), this becomes

$$\mathfrak{F}_N(\tau) = \frac{1}{N}\left|e^{i2\pi\tau L_N\gamma_1} + \sum_{\substack{1\le m\le N-1\\ m\ \text{odd}}} \sqrt{\frac{2}{N}}\cot\left(\frac{\pi m}{2N}\right) A_m(N;2\pi\tau L_N)\right|^2. \tag{6.237}$$

Thus the Montgomery form factor is exactly the squared norm of the constant Green-modal projection.

The same $F_N(\tau)$ has the standard Montgomery double-sum form. Let

$$w(x) = \frac{4}{4 + x^2}. \tag{6.238}$$

We use the Fourier identity

$$w(x) = \int_{-\infty}^{\infty} e^{-2|y|} e^{ixy}\, dy. \tag{6.239}$$

By (6.239), the endpoint-integral definition (6.223) is equivalent to

$$F_N(\tau) = \frac{1}{N} \sum_{j,k=1}^{N} (T_N/2\pi)^{i\tau(\gamma_j - \gamma_k)} w(\gamma_j - \gamma_k). \tag{6.240}$$

Using the endpoint projection identity,

$$F_N(\tau) = \frac{1}{N} \int_{-\infty}^{\infty} e^{-2|y|} \left| e^{i(\tau \log(T_N/2\pi) + y)\gamma_1} + \sum_{\substack{1 \le m \le N-1 \\ m \text{ odd}}} u_m A_m(N; \tau \log(T_N/2\pi) + y) \right|^2 dy. \tag{6.241}$$

This is exact for every N. It is the smoothed Montgomery form factor written entirely in Green-modal coordinates.

Proposition 6.29 (Finite-to-limit equivalence)**.** *For each N, identity* (6.241) *is exact. Consequently, the limiting assertion*

$$F_N(\tau) \to \min(|\tau|, 1) \tag{6.242}$$

is equivalent to convergence of the corresponding Green-modal endpoint projection

$$e^{it\gamma_1} + u^* A(N; t), \qquad t = \tau \log(T_N/2\pi) + y. \tag{6.243}$$

The elementary modal bounds above give uniform finite-N control. The finite Green deweighting theorem gives one-sided endpoint control from the Green energy, as recorded in Corollary 6.28. The Montgomery–GUE target is then an asymptotic statement about the endpoint projection, or equivalently about the modal expression in (6.241)*.*

6.10. **From the deformed phase walk to the sine kernel.** We reserve j, k, n, N for ordinal indices of zeros, while T, T_N, and T_0 denote heights. For a prefix of length N, we write

$$T_N = \gamma_N. \tag{6.244}$$

For a local window, T_0 denotes a representative height of the window, for instance its central or mean height. Thus the Montgomery scaling is

$$t = \tau \log(T_N/2\pi) \tag{6.245}$$

for prefixes and

$$t = \tau \log(T_0/2\pi) \tag{6.246}$$

for local windows.

The original phase walk is the fixed section of a deformed family:

$$S_N = S_N(1) = \sum_{k=1}^{N} e^{i\gamma_k}, \qquad S_N(t) = \sum_{k=1}^{N} e^{it\gamma_k}. \tag{6.247}$$

The section $t = 1$ corresponds to the fixed real frequency $X = e$. Its role in the present paper is the fixed-frequency non-resonant confinement supplied by Landau's formula, together with the associated centroid and Green-weighted centered moment. The Montgomery observable belongs to the full deformed family $S_N(t)$, in the local scaling (6.245) or (6.246).

Squaring the deformed endpoint gives all ordered pairs:

$$\begin{aligned} |S_N(t)|^2 &= \left(\sum_{j\le N} e^{it\gamma_j}\right)\left(\sum_{k\le N} e^{-it\gamma_k}\right) \\ &= \sum_{j,k\le N} e^{it(\gamma_j-\gamma_k)}. \end{aligned} \tag{6.248}$$

Thus the Montgomery form factor is a statistic of pair differences $\gamma_j - \gamma_k$, not only of consecutive gaps.

For a local window of length W, centered near height T_0, define the unfolded pair variable

$$\xi_{jk} = \frac{\log(T_0/2\pi)}{2\pi}(\gamma_j - \gamma_k). \tag{6.249}$$

With $t = \tau \log(T_0/2\pi)$, one has

$$e^{it(\gamma_j-\gamma_k)} = e^{i2\pi\tau\xi_{jk}}. \tag{6.250}$$

Consequently, for the window endpoint,

$$\frac{1}{W}\,|S_W(\tau\log(T_0/2\pi))|^2 = \frac{1}{W}\sum_{j,k\le W} e^{i2\pi\tau\xi_{jk}}. \tag{6.251}$$

Thus $S_W(t)$ gives the Fourier transform of the unfolded pair correlation attached to the local window. In Fourier variables the Montgomery–GUE target is the form factor

$$F(\tau) = \min(|\tau|, 1), \tag{6.252}$$

while in unfolded pair variables it is the sine-kernel pair correlation

$$R_2(\xi) = 1 - \left(\frac{\sin\pi\xi}{\pi\xi}\right)^2. \tag{6.253}$$

These are dual descriptions of the same pair-correlation structure: the ramp-plateau is the Fourier-side form factor, and the sine kernel is the side of unfolded differences.

The endpoint reconstruction identity gives the same statement in Green-modal coordinates:

$$S_N(t) = e^{it\gamma_1} + \sum_{\substack{1\le m\le N-1 \\ m \text{ odd}}} u_m A_m(N;t). \tag{6.254}$$

Hence the smoothed Montgomery form factor is

$$F_N(\tau) = \frac{1}{N}\int_{-\infty}^{\infty} e^{-2|y|}\left|e^{i(\tau\log(T_N/2\pi)+y)\gamma_1} + \sum_{\substack{1\le m\le N-1\\ m\ \text{odd}}} u_m A_m(N;\tau\log(T_N/2\pi)+y)\right|^2 dy. \tag{6.255}$$

This uses the standard local normalization $\log(T_N/2\pi)$, consistent with the Riemann–von Mangoldt mean spacing.

The Green energy and the Montgomery endpoint projection are different outputs of the same vector $A(N;t)$. The Green-energy output is

$$A(N;t) \longmapsto A(N;t)^*\Lambda_N A(N;t) = \sum_{m=1}^{N-1}\lambda_m |A_m(N;t)|^2, \tag{6.256}$$

which is the t-deformed centered Green moment. The endpoint output is

$$A(N;t) \longmapsto \frac{1}{N}\left|e^{it\gamma_1} + u^*A(N;t)\right|^2, \tag{6.257}$$

which is the Montgomery form-factor observable. The Green energy $A^*\Lambda A$ and the Montgomery endpoint projection $N^{-1}|e^{it\gamma_1}+u^*A|^2$ are distinct observables. They are two different outputs of the same modal vector $A(N;t)$.

This framework separates the fixed-frequency confinement of $S_N(1)$ from the Montgomery–GUE behaviour encoded by the deformed family $S_N(t)$. The Fourier-side form factor and the unfolded pair-side sine-kernel are two dual descriptions of the same pair-correlation structure.

6.11. **Finite modal identities and the scaled modal variable.**

Lemma 6.30 (Finite modal integration)**.** *For each fixed N, each fixed τ, and the smoothed Montgomery parameter $t_N(\tau,y) = \tau\log(T_N/2\pi)+y$, the y-integral in the smoothed form factor may be interchanged with the finite modal sums in* (6.241)*. More explicitly,*

$$\int_{-\infty}^{\infty} e^{-2|y|}\left|e^{it_N(\tau,y)\gamma_1} + \sum_{\substack{1\le m\le N-1\\ m\ \text{odd}}} u_m A_m(N;t_N(\tau,y))\right|^2 dy \tag{6.258}$$

is the integral of a finite sum of continuous functions of y, and the interchange is justified by finite linearity of the integral. No interchange of an infinite modal sum with the limit $N\to\infty$ is used.

Proof. For fixed N, the modal expansion contains at most $N-1$ terms. For each m, the coefficient $A_m(N;t_N(\tau,y))$ is a finite sum of exponentials in y. Hence it is continuous and bounded by (6.129). Since $\int_{\mathbb{R}} e^{-2|y|}\,dy = 1$, every term is integrable against the smoothing weight. The desired interchange is therefore only the linearity of integration over a finite sum. □

Lemma 6.31 (Smoothed modal L^2 control)**.** *For every fixed N, τ, and $1\le m\le N-1$,*

$$\int_{-\infty}^{\infty} e^{-2|y|}|A_m(N;t_N(\tau,y))|^2\,dy \le N-1. \tag{6.259}$$

Moreover,

$$\int_{-\infty}^{\infty} e^{-2|y|}\sum_{m=1}^{N-1}|A_m(N;t_N(\tau,y))|^2\,dy = N-1. \tag{6.260}$$

Proof. The first bound is (6.139). For the second identity, use Parseval (6.130) pointwise in $t = t_N(\tau, y)$, then integrate against $e^{-2|y|}$. The exchange of the finite sum over m with the y-integral is justified by Lemma 6.30. □

The Green-modal decomposition is a finite-dimensional identity. For each fixed N,

$$\mathcal{E}_N(t) = A(N;t)^* \Lambda_N A(N;t) = \sum_{m=1}^{N-1} \lambda_m |A_m(N;t)|^2 \tag{6.261}$$

contains only $N-1$ terms. Hence no convergence issue is involved in the identity itself. Lemma 6.30 records the only exact interchange between y-integration and modal summation used below, while Lemma 6.31 records the finite-N L^2 domination used in the smoothed covariance estimates.

The passage $N \to \infty$ is a different question. In the Montgomery scaling

$$t = \tau \log(T_N/2\pi) + y, \tag{6.262}$$

the local zero spacing satisfies heuristically

$$\gamma_{k+1} - \gamma_k \sim \frac{2\pi}{\log(T_N/2\pi)}. \tag{6.263}$$

Thus

$$t(\gamma_{k+1} - \gamma_k) \sim 2\pi\tau. \tag{6.264}$$

Consequently, as a function of the index k, the vector

$$e^{it\gamma_k} \tag{6.265}$$

has macroscopic oscillation. In the sine basis

$$\phi_m(r) = \sqrt{\frac{2}{N}} \sin\left(\frac{\pi m r}{N}\right), \tag{6.266}$$

such oscillations are naturally represented by modal indices of order N, not by finitely many fixed modes $m = O(1)$.

Therefore a low-mode truncation $m \le M$ with fixed M is not a natural approximation to the Montgomery endpoint projection. Any genuine modal limiting theorem in the Montgomery regime should be formulated using the scaled modal variable

$$\theta = \frac{m}{N} \tag{6.267}$$

or an equivalent frequency-window formulation.

The unconditional limiting statement obtained in the present paper is instead the finite Green deweighting consequence

$$F_N(\tau) \le 2(1+\eta) \int_{-\infty}^{\infty} e^{-2|y|} \mathcal{E}_N(\tau \log(T_N/2\pi) + y)\, dy + \frac{1+\eta^{-1}}{N}, \tag{6.268}$$

and hence

$$\limsup_{N\to\infty} F_N(\tau) \le 2 \limsup_{N\to\infty} \int_{-\infty}^{\infty} e^{-2|y|} \mathcal{E}_N(\tau \log(T_N/2\pi) + y)\, dy. \tag{6.269}$$

This uses only finite-N identities and finite-N inequalities; it does not require an unproved interchange of an infinite modal sum with the limit $N \to \infty$.

6.12. **Smoothed modal covariance and conditional form-factor criteria.** The pointwise bound (6.129) and the Parseval identity (6.130) are unconditional finite-dimensional estimates. They are useful for normalization, but they do not by themselves determine the Montgomery form factor. The form factor depends on the correlated modal contraction in the constant-mode direction.

For fixed τ, put

$$t_N(\tau, y) = \tau \log(T_N/2\pi) + y. \tag{6.270}$$

Define the smoothed modal covariance matrix

$$\Sigma_N(\tau) = \int_{-\infty}^{\infty} e^{-2|y|} A(N; t_N(\tau, y)) A(N; t_N(\tau, y))^*\, dy. \tag{6.271}$$

Thus

$$\Sigma_N(\tau)_{m\ell} = \int_{-\infty}^{\infty} e^{-2|y|} A_m(N; t_N(\tau, y)) \overline{A_\ell(N; t_N(\tau, y))}\, dy. \tag{6.272}$$

Proposition 6.32 (Exact modal covariance expression for F_N)**.** *One has*

$$\begin{aligned} F_N(\tau) = \frac{1}{N} + \frac{2}{N} \Re \int_{-\infty}^{\infty} e^{-2|y|} e^{-i t_N(\tau,y)\gamma_1} u^* A(N; t_N(\tau, y))\, dy \\ + \frac{1}{N} u^* \Sigma_N(\tau) u. \end{aligned} \tag{6.273}$$

Moreover, if

$$\frac{1}{N} u^* \Sigma_N(\tau) u = O(1), \tag{6.274}$$

then the middle term in (6.273) *is* $o(1)$*, and therefore*

$$F_N(\tau) = \frac{1}{N} u^* \Sigma_N(\tau) u + o(1). \tag{6.275}$$

Proof. Starting from (6.241) and using $t = t_N(\tau, y)$, expand the square. The first term gives

$$\frac{1}{N} \int_{-\infty}^{\infty} e^{-2|y|}\, dy = \frac{1}{N}. \tag{6.276}$$

The quadratic modal term is exactly

$$\frac{1}{N} u^* \left(\int_{-\infty}^{\infty} e^{-2|y|} A(N; t_N(\tau, y)) A(N; t_N(\tau, y))^*\, dy \right) u = \frac{1}{N} u^* \Sigma_N(\tau) u. \tag{6.277}$$

This gives (6.273). For the middle term, Cauchy–Schwarz gives

$$\begin{aligned}&\left|\int_{-\infty}^{\infty} e^{-2|y|}e^{-it_N(\tau,y)\gamma_1}u^*A(N;t_N(\tau,y))\,dy\right|^2\\&\qquad\le\left(\int_{-\infty}^{\infty}e^{-2|y|}\,dy\right)\left(\int_{-\infty}^{\infty}e^{-2|y|}|u^*A(N;t_N(\tau,y))|^2\,dy\right)\\&\qquad= u^*\Sigma_N(\tau)u.\end{aligned}\tag{6.278}$$

Hence the middle term is bounded by

$$\frac{2}{N}\left(u^*\Sigma_N(\tau)u\right)^{1/2}.\tag{6.279}$$

Under (6.274), this is $O(N^{-1/2})=o(1)$. This proves (6.275). □

Conjecture 6.33 (Modal covariance form of Montgomery)**.** *For each fixed $\tau\in\mathbb{R}$, the smoothed modal covariance contraction satisfies*

$$\frac{1}{N}u^*\Sigma_N(\tau)u\longrightarrow\min(|\tau|,1).\tag{6.280}$$

Corollary 6.34 (Modal covariance criterion for the Montgomery form factor)**.** *Fix $\tau\in\mathbb{R}$. All convergence in this corollary concerns the smoothed y-averaged form factor $F_N(\tau)$ defined in* (6.223)*, pointwise in this fixed value of τ. Suppose that*

$$\frac{1}{N}u^*\Sigma_N(\tau)u\longrightarrow L(\tau)\tag{6.281}$$

for a finite limit $L(\tau)$. Then

$$F_N(\tau)\longrightarrow L(\tau).\tag{6.282}$$

In particular, the Montgomery–GUE form-factor target is equivalent, in this modal language, to Conjecture 6.33, namely to the conditional modal covariance contraction

$$\frac{1}{N}u^*\Sigma_N(\tau)u\longrightarrow\min(|\tau|,1).\tag{6.283}$$

Proof. The convergence assumption (6.281) implies the boundedness condition (6.274). The conclusion follows from Proposition 6.32. □

Remark 6.35 (Individual modal bounds versus modal covariance)*.* The estimates (6.129) and (6.130) are unconditional. They do not determine the Montgomery form factor, because $F_N(\tau)$ depends on the rank-one modal contraction $u^*\Sigma_N(\tau)u$, which includes correlations between different modes. Thus individual modal bounds provide finite control, while the limiting form factor is governed by the smoothed modal covariance in the constant-mode direction u.

The Green energy gives a useful one-sided consequence. Since (6.186) gives

$$|u^*A(N;t)|^2\le 2N\,\mathcal{E}_N(t),\tag{6.284}$$

one has

$$\frac{1}{N}u^*\Sigma_N(\tau)u=\frac{1}{N}\int_{-\infty}^{\infty}e^{-2|y|}|u^*A(N;t_N(\tau,y))|^2\,dy\le 2\int_{-\infty}^{\infty}e^{-2|y|}\mathcal{E}_N(t_N(\tau,y))\,dy.\tag{6.285}$$

Consequently, a bound on the smoothed Green energy gives a bound on the modal covariance contraction and hence on $F_N(\tau)$, but not exact recovery of the limiting form factor unless the covariance contraction itself is controlled.

The Montgomery–GUE target in this notation is

$$F_N(\tau) \longrightarrow \min(|\tau|, 1). \tag{6.286}$$

Equivalently, the Green-modal target is

$$\frac{1}{N}\int_{-\infty}^{\infty} e^{-2|y|}\left|e^{i(\tau\log(T_N/2\pi)+y)\gamma_1} + \sum_{\substack{1\le m\le N-1\\ m \text{ odd}}} u_m A_m(N;\tau\log(T_N/2\pi)+y)\right|^2 dy \longrightarrow \min(|\tau|,1). \tag{6.287}$$

This is the Green-modal form of the Montgomery form-factor conjecture.

Remark 6.36 (Fixed frequencies versus Montgomery scaling). Landau fixed-frequency formulas concern fixed frequencies. The non-integer fixed frequency used in the complete-zero walk is covered by Proposition 6.2; at $X = e$ it gives

$$\sum_{0<\gamma\le T} e^{\rho} = O(\log T). \tag{6.288}$$

For integral prime-power frequencies the fixed-frequency Landau–Gonek main term has negative sign and is supported at the arithmetic resonances $X = p^m$. In the fixed-real notation $\Lambda_{\mathbb{R}}(e) = 0$, since e is not an integral prime power.

Under RH, the $X = e$ formula gives the corresponding fixed-frequency phase statement

$$\sum_{0<\gamma\le T} e^{i\gamma} = O(\log T). \tag{6.289}$$

Montgomery's form factor belongs to a different regime. In the present ordinal normalization the frequency is N-dependent; one takes

$$X_N = (T_N/2\pi)^{\tau}, \qquad t = \tau\log(T_N/2\pi), \tag{6.290}$$

After squaring and smoothing, the diagonal prime contributions in the explicit formula accumulate over the growing range

$$n \le (T_N/2\pi)^{|\tau|}. \tag{6.291}$$

At the formal explicit-formula level, this is the standard ramp mechanism

$$F_N(\tau) \sim \frac{1}{\log T_N}\sum_{n\le(T_N/2\pi)^{|\tau|}} \frac{\Lambda(n)^2}{n} \sim |\tau|, \qquad 0 < |\tau| < 1. \tag{6.292}$$

The fixed frequency $X = e$ is non-resonant for the phase walk S_N, whereas the Montgomery scaling $X_N = (T_N/2\pi)^{\tau}$ explores a growing arithmetic range after squaring and smoothing.

Thus the form-factor target has ramp

$$F_N(\tau) \to |\tau|, \qquad 0 < |\tau| < 1. \tag{6.293}$$

and saturation

$$F_N(\tau) \to 1, \qquad |\tau| \geq 1. \tag{6.294}$$

Together these give

$$F_N(\tau) \to \min(|\tau|, 1). \tag{6.295}$$

Finally connect this to the sine kernel. With the convention

$$\widehat{f}(\tau) = \int_{\mathbb{R}} f(u) e^{-2\pi i \tau u}\, du, \tag{6.296}$$

one has

$$\left(\widehat{\frac{\sin \pi u}{\pi u}}\right)^2 = (1 - |\tau|)_+. \tag{6.297}$$

Since

$$\min(|\tau|, 1) - 1 = -(1 - |\tau|)_+, \tag{6.298}$$

the corresponding limiting pair-correlation density is

$$R_2(u) = 1 - \left(\frac{\sin \pi u}{\pi u}\right)^2. \tag{6.299}$$

The Green/Brownian framework does not merely rewrite the centered moment $M_0(N;t)$. It also supplies a canonical modal representation of the Montgomery form factor. The Green energy is the diagonal quadratic form

$$A(N;t)^* \Lambda_N A(N;t), \tag{6.300}$$

whereas the Montgomery form factor is the rank-one constant-mode contraction

$$\frac{1}{N} \left| e^{it\gamma_1} + u^* A(N;t) \right|^2. \tag{6.301}$$

Thus the same Green-modal vector $A(N;t)$ carries both the centered phase-walk energy and the pair-correlation form factor. In this language, the Montgomery–GUE form-factor target is the limiting energy $\min(|\tau|, 1)$ of the constant-mode Green projection in the unfolded scaling

$$t = \tau \log(T_N/2\pi). \tag{6.302}$$

6.13. **Riemann–von Mangoldt background and modal residuals.** The Green-modal representation also separates a smooth counting background from the genuine zero fluctuations. Define the smooth Riemann–von Mangoldt main term

$$N_0(T) = \frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + \frac{7}{8}. \tag{6.303}$$

We use RVM as shorthand for Riemann–von Mangoldt in the notation below. The corresponding RVM grid is the deterministic grid obtained by inverting this smooth counting function.

Let

$$\widetilde{\gamma}_k = N_0^{-1}(k), \tag{6.304}$$

and define the residual displacement

$$d_k = \gamma_k - \widetilde{\gamma}_k. \tag{6.305}$$

The modal coefficient decomposes as

$$A_m(N;t) = A_m^{\mathrm{RVM}}(N;t) + R_m(N;t), \tag{6.306}$$

where

$$A_m^{\mathrm{RVM}}(N;t) = \sum_{r=1}^{N-1} \phi_m(r) e^{it\widetilde{\gamma}_{N-r+1}}, \tag{6.307}$$

and

$$R_m(N;t) = \sum_{r=1}^{N-1} \phi_m(r) e^{it\widetilde{\gamma}_{N-r+1}} \left(e^{itd_{N-r+1}} - 1\right). \tag{6.308}$$

Lemma 6.37 (Parseval bound for the RVM modal residual)**.** *Let*

$$\widetilde{\gamma}_k = N_0^{-1}(k), \qquad d_k = \gamma_k - \widetilde{\gamma}_k, \tag{6.309}$$

and define $R_m(N;t)$ by (6.308)*. Then*

$$\sum_{m=1}^{N-1} |R_m(N;t)|^2 = \sum_{k=2}^{N} |e^{itd_k} - 1|^2. \tag{6.310}$$

Consequently,

$$\sum_{m=1}^{N-1} |R_m(N;t)|^2 \le 4(N-1). \tag{6.311}$$

Moreover,

$$\sum_{m=1}^{N-1} |R_m(N;t)|^2 \le t^2 \sum_{k=2}^{N} |d_k|^2. \tag{6.312}$$

Proof. Set

$$b_r(t) = e^{it\widetilde{\gamma}_{N-r+1}}\left(e^{itd_{N-r+1}} - 1\right), \qquad 1 \le r \le N-1. \tag{6.313}$$

Then

$$R_m(N;t) = \sum_{r=1}^{N-1} \phi_m(r) b_r(t). \tag{6.314}$$

Since

$$\phi_m(r) = \sqrt{\frac{2}{N}} \sin\left(\frac{\pi m r}{N}\right), \qquad 1 \le m, r \le N-1, \tag{6.315}$$

is an orthonormal sine basis of $\mathbb{C}^{N-1}$, Parseval's identity gives

$$\sum_{m=1}^{N-1} |R_m(N;t)|^2 = \sum_{r=1}^{N-1} |b_r(t)|^2. \tag{6.316}$$

Because

$$|e^{it\widetilde{\gamma}_{N-r+1}}| = 1, \tag{6.317}$$

we have

$$|b_r(t)|^2 = \left|e^{itd_{N-r+1}} - 1\right|^2. \tag{6.318}$$

Changing variables $k = N - r + 1$, this becomes

$$\sum_{m=1}^{N-1} |R_m(N;t)|^2 = \sum_{k=2}^{N} |e^{itd_k} - 1|^2. \tag{6.319}$$

The bound

$$|e^{ix} - 1| \le 2 \tag{6.320}$$

gives

$$\sum_{m=1}^{N-1} |R_m(N;t)|^2 \le 4(N-1). \tag{6.321}$$

The sharper small-displacement estimate follows from

$$|e^{ix} - 1| \le |x|, \tag{6.322}$$

which gives

$$|e^{itd_k} - 1|^2 \le t^2 |d_k|^2. \tag{6.323}$$

Summing over k proves the final bound. □

The estimate $4(N-1)$ is a universal Parseval bound. It is sharp at this level of information: without further control on the phases td_k, the individual quantities $|e^{itd_k}-1|$ may be as large as 2, so no $o(N)$ improvement follows from orthogonality alone.

For fixed t, the refinement

$$\sum_{m=1}^{N-1} |R_m(N;t)|^2 \le t^2 \sum_{k=2}^{N} |d_k|^2 \tag{6.324}$$

can be stronger if the displacements $d_k = \gamma_k - \widetilde{\gamma}_k$ are square-summably small on average. In Montgomery scaling, however,

$$t = \tau \log(T_N/2\pi) + y, \tag{6.325}$$

and the products td_k need not be small. Thus the Parseval estimate gives uniform modal boundedness after normalization,

$$\frac{1}{N} \sum_{m=1}^{N-1} |R_m(N;t)|^2 \le 4, \tag{6.326}$$

and it shows that the RVM residual has controlled total modal energy.

Thus A_m^{RVM} is the calculable smooth background, while R_m is the modal remainder caused by replacing the synthetic Riemann–von Mangoldt grid ordinates $\widetilde{\gamma}_k$ with the actual zero ordinates γ_k. The RVM grid alone is rigid after unfolding; it is not a GUE process. The Green-modal form factor separates this smooth background from the residual fluctuations that carry the arithmetic and spectral information of the actual zeros.

7. Supplementary Exact Product Forms for S_N and C_N

This supplementary section gives exact algebraic reformulations of the cumulative sum S_N and of the centroid C_N in terms of finite products associated with the phases $e^{i\gamma_k}$. These identities are complementary algebraic structure; the main structural result of the paper remains the Brownian-bridge/Green-kernel formulation developed above.

7.1. **A Finite Product Identity.**

We first record the elementary identity on which the product formulation is based.

Lemma 7.1 (Logarithmic coefficients of a finite product)**.** *Let $a_1, \dots, a_N$ be complex numbers, and define*

$$P(z) = \prod_{k=1}^{N} (1 - a_k z). \tag{7.1}$$

For $|z|$ sufficiently small,

$$\log P(z) = -\sum_{m \ge 1} \frac{z^m}{m} \sum_{k=1}^{N} a_k^m. \tag{7.2}$$

In particular,

$$-\frac{d}{dz} \log P(z)\Big|_{z=0} = \sum_{k=1}^{N} a_k. \tag{7.3}$$

Proof. Since $P(z)$ is a finite product,

$$\log P(z) = \sum_{k=1}^{N} \log(1 - a_k z). \tag{7.4}$$

Using

$$\log(1 - w) = -\sum_{m \geq 1} \frac{w^m}{m} \tag{7.5}$$

for $|w|$ sufficiently small, we obtain

$$\log P(z) = -\sum_{k=1}^{N} \sum_{m \geq 1} \frac{(a_k z)^m}{m} \tag{7.6}$$

$$= -\sum_{m \geq 1} \frac{z^m}{m} \sum_{k=1}^{N} a_k^m. \tag{7.7}$$

Differentiating at $z = 0$ gives (7.3).

□

7.2. **The Phase Product and the Sum S_N.** We now apply Lemma 7.1 to the phases

$$a_k = e^{i\gamma_k}. \tag{7.8}$$

Definition 7.2 (Phase product)**.** The phase product associated with the first N ordinates is

$$P_N(z) = \prod_{k=1}^{N} \left(1 - z e^{i\gamma_k}\right). \tag{7.9}$$

The zeros of P_N are

$$z = e^{-i\gamma_k}, \qquad k = 1, \ldots, N. \tag{7.10}$$

Thus all zeros of P_N lie on the unit circle.
For each integer $m \geq 1$, define the m-th phase sum by

$$S_N^{(m)} = \sum_{k=1}^{N} e^{im\gamma_k}. \tag{7.11}$$

Then Lemma 7.1 gives

$$\log P_N(z) = -\sum_{m \geq 1} \frac{S_N^{(m)}}{m} z^m. \tag{7.12}$$

In particular,

$$S_N = S_N^{(1)}. \tag{7.13}$$

Equivalently,

$$S_N = -\frac{d}{dz}\log P_N(z)\Big|_{z=0}. \tag{7.14}$$

Thus S_N is the first logarithmic coefficient of the phase product P_N.

7.3. **Polynomial Form and Elementary Symmetric Coefficients.** It is also useful to consider the monic polynomial

$$Q_N(z) = \prod_{k=1}^{N}\left(z - e^{i\gamma_k}\right). \tag{7.15}$$

Expanding Q_N, one obtains

$$Q_N(z) = z^N - e_1^{(N)} z^{N-1} + e_2^{(N)} z^{N-2} - \cdots + (-1)^N e_N^{(N)}. \tag{7.16}$$

Here

$$e_r^{(N)} = \sum_{1\le k_1<\cdots<k_r\le N} e^{i(\gamma_{k_1}+\cdots+\gamma_{k_r})}. \tag{7.17}$$

In particular,

$$e_1^{(N)} = \sum_{k=1}^{N} e^{i\gamma_k} = S_N. \tag{7.18}$$

Hence

$$Q_N(z) = z^N - S_N z^{N-1} + e_2^{(N)} z^{N-2} - \cdots + (-1)^N e_N^{(N)}. \tag{7.19}$$

This identifies S_N as the first elementary symmetric coefficient of the phases $e^{i\gamma_k}$.

7.4. **Product Form of the Centroid.** The centroid was defined by

$$C_N = \frac{1}{N}\sum_{n=1}^{N} S_n. \tag{7.20}$$

For each $n \le N$, define the partial phase product

$$P_n(z) = \prod_{k=1}^{n}\left(1 - z e^{i\gamma_k}\right). \tag{7.21}$$

By (7.14), applied with N replaced by n, one has

$$S_n = -\frac{d}{dz}\log P_n(z)\Big|_{z=0}. \tag{7.22}$$

Substitution into (7.20) gives

$$C_N = -\frac{1}{N}\sum_{n=1}^{N}\frac{d}{dz}\log P_n(z)\Big|_{z=0}. \tag{7.23}$$

By linearity of differentiation,

$$C_N = -\frac{d}{dz}\left(\frac{1}{N}\sum_{n=1}^{N}\log P_n(z)\right)\Big|_{z=0}. \tag{7.24}$$

Definition 7.3 (Accumulated phase product)**.** The accumulated phase product is defined by

$$\mathcal{P}_N(z) = \prod_{n=1}^{N} P_n(z). \tag{7.25}$$

Since

$$\log \mathcal{P}_N(z) = \sum_{n=1}^{N}\log P_n(z), \tag{7.26}$$

we obtain the exact product identity

$$C_N = -\frac{1}{N}\,\frac{d}{dz}\log \mathcal{P}_N(z)\Big|_{z=0}. \tag{7.27}$$

Thus C_N is the normalized first logarithmic coefficient of the accumulated phase product $\mathcal{P}_N$.

7.5. **The Triangular Product.** Using the definition of P_n,

$$\mathcal{P}_N(z) = \prod_{n=1}^{N}\prod_{k=1}^{n}\left(1 - ze^{i\gamma_k}\right). \tag{7.28}$$

Changing the order of multiplication gives

$$\mathcal{P}_N(z) = \prod_{k=1}^{N}\left(1 - ze^{i\gamma_k}\right)^{N-k+1}. \tag{7.29}$$

Combining (7.27) and (7.29), one obtains

$$C_N = -\frac{1}{N}\,\frac{d}{dz}\log\prod_{k=1}^{N}\left(1 - ze^{i\gamma_k}\right)^{N-k+1}\Big|_{z=0}. \tag{7.30}$$

Differentiating explicitly gives

$$C_N = \frac{1}{N}\sum_{k=1}^{N}(N-k+1)e^{i\gamma_k}. \tag{7.31}$$

This is the triangular weighted form of the centroid.

7.6. **Triangular Harmonic Interpretation.** The usual empirical measure of the phases is

$$\nu_N = \frac{1}{N}\sum_{k=1}^{N}\delta_{e^{i\gamma_k}}. \tag{7.32}$$

Its first moment is

$$\int_{\mathbb{T}} z\, d\nu_N(z) = \frac{1}{N}\sum_{k=1}^{N} e^{i\gamma_k} = \frac{S_N}{N}. \tag{7.33}$$

The centroid C_N, by contrast, corresponds to the triangularly weighted finite measure

$$\mu_N^{\triangle} = \frac{1}{N}\sum_{k=1}^{N}(N-k+1)\delta_{e^{i\gamma_k}}. \tag{7.34}$$

Its first moment is

$$\int_{\mathbb{T}} z\, d\mu_N^{\triangle}(z) = \frac{1}{N}\sum_{k=1}^{N}(N-k+1)e^{i\gamma_k} = C_N. \tag{7.35}$$

Therefore C_N is the first harmonic of a triangularly weighted phase measure, whereas S_N/N is the first harmonic of the uniformly weighted empirical phase measure.

7.7. **Local Analytic Interpretation at the Origin.** From (7.12),

$$\log P_N(z) = -S_N z + O(z^2) \qquad (z \to 0). \tag{7.36}$$

Thus

$$S_N = -\left.\frac{d}{dz}\log P_N(z)\right|_{z=0}. \tag{7.37}$$

Similarly, (7.27) identifies C_N as the normalized local logarithmic derivative of the accumulated phase product $\mathcal{P}_N$ at the origin.

Consequently, S_N and C_N are local logarithmic derivatives at the origin of two finite products associated with the phases $e^{i\gamma_k}$: the ordinary phase product P_N and the accumulated phase product $\mathcal{P}_N$, respectively.

Remark 7.4. The identities above are exact finite-N identities.

They do not depend on the Riemann Hypothesis, on random matrix models, or on probabilistic assumptions about the zeros.

They show that S_N is the first logarithmic coefficient of the ordinary phase product P_N, while C_N is the normalized first logarithmic coefficient of the accumulated product $\mathcal{P}_N$.

8. Supplementary Non-Resonant Triangular Landau–Gonek Structure of S_N and C_N

This supplementary section records rigorous structural identities for the ordinal sum S_N, the height-truncated sums $S_X(T)$, and the centroid C_N. The Landau–Gonek phase formulas recalled here are literature-based; the terminology describing C_N as a triangular non-resonant transform is an original structural interpretation of the identities below.

8.1. **Ordinal and height truncations.** The sum S_N is an ordinal truncation: N is a positive integer and counts the first N ordinates in the ordered list (3.3). By contrast, height truncations are cut off by an analytic height T on the ordinate axis. The counting function $N_\zeta(T)$, defined in (2.14), translates the height T into the corresponding ordinal number of zeros. Thus N is not identified with T.

Definition 8.1. For $X \in \mathbb{R}_{>0}$ and $T > 0$, define

$$S_X(T) = \sum_{0<\gamma\le T} X^{i\gamma}, \tag{8.1}$$

where zeros are counted with multiplicity and

$$X^{i\gamma} := \exp(i\gamma \log X) \tag{8.2}$$

with the real logarithm of X.

Theorem 8.2 (Height-to-ordinal bridge)**.** *For every $T > 0$,*

$$S_e(T) = \sum_{0<\gamma\le T} e^{i\gamma} = \sum_{k=1}^{N_\zeta(T)} e^{i\gamma_k} = S_{N_\zeta(T)}. \tag{8.3}$$

Proof. The ordered sequence (3.3) lists the positive ordinates of the nontrivial zeros with multiplicity. By definition, $N_\zeta(T)$ is exactly the number of entries in this list whose ordinate is at most T. Therefore the height-truncated multiset $\{\gamma : 0 < \gamma \le T\}$, counted with multiplicity, is the same multiset as $\{\gamma_k : 1 \le k \le N_\zeta(T)\}$. Summing $e^{i\gamma}$ over these two identical multisets gives the first two equalities in (8.3). The last equality is precisely the ordinal definition (5.2) with $N = N_\zeta(T)$. □

Corollary 8.3 (Ordinal recovery away from ambiguity)**.** *For every height $T > 0$ such that $N_\zeta(T) = N$,*

$$S_N = S_e(T). \tag{8.4}$$

Proof. Substitute the hypothesis $N_\zeta(T) = N$ into (8.3). □

8.2. **Continuous phase sums.** The parameter X in (8.1) is a positive real continuous frequency. The definition (8.2) avoids any multivalued complex logarithm.

Lemma 8.4 (The continuous frequency $X = e$)**.** *At the continuous frequency $X = e$, the height-truncated phase sum is*

$$S_e(T) = \sum_{0<\gamma\le T} e^{i\gamma}. \tag{8.5}$$

Proof. By the definition of the continuous power with positive real base, for every $X \in \mathbb{R}_{>0}$, the identity

$$X^{i\gamma} = \exp(i\gamma \log X). \tag{8.6}$$

holds. Taking $X = e$, and using $\log e = 1$, gives

$$X^{i\gamma}\big|_{X=e} = e^{i\gamma} = \exp(i\gamma). \tag{8.7}$$

Substituting this identity into the definition of $S_X(T)$ with $X = e$ proves the claim. □

Remark 8.5. The frequency $X = e$ is a fixed positive real frequency. In the arithmetic Landau–Gonek formula, resonances occur at integral prime powers. With the extended convention

$$\Lambda_{\mathbb{R}}(X) = 0 \tag{8.8}$$

away from prime powers, one has

$$\Lambda_{\mathbb{R}}(e) = 0. \tag{8.9}$$

Equivalently, e is non-resonant because it is not an integral prime power.

8.3. **Arithmetic Landau–Gonek resonances.** Landau–Gonek resonances are arithmetic. In this subsection $X \in \mathbb{R}_{>0}$ denotes a continuous frequency, while $x \in \mathbb{N}$, $x > 1$, denotes an arithmetic frequency.

Definition 8.6. The arithmetic resonance set is

$$\mathcal{R} = \{p^m : \ p \text{ prime}, \ m \in \mathbb{N}\}. \tag{8.10}$$

For fixed $x \in \mathbb{N}$, $x > 1$, the Landau–Gonek formula under RH has the phase form

$$\sum_{0<\gamma\le T} x^{i\gamma} = -\frac{T}{2\pi}\frac{\Lambda(x)}{\sqrt{x}} + O(\log T). \tag{8.11}$$

The main term is supported on the prime-power set $\mathcal{R}$.

Theorem 8.7 (Non-resonance of e)**.** *The continuous frequency $X = e$ is not an arithmetic Landau–Gonek resonance:*

$$e \notin \mathcal{R}. \tag{8.12}$$

Proof. Every element of $\mathcal{R}$ is a positive integer, so $\mathcal{R} \subset \mathbb{N}$. The number e is irrational, hence $e \notin \mathbb{N}$. Therefore $e \notin \mathcal{R}$. □

Remark 8.8. In the extended real-frequency notation, $\Lambda_{\mathbb{R}}(X) = 0$ away from integral prime powers. Thus $\Lambda_{\mathbb{R}}(e) = 0$, equivalently $X = e$ is non-resonant because it is not an integral prime power.

8.4. **The centroid as a triangular non-resonant transform.** The centroid identity is purely ordinal and contains no height variable. It follows from the definitions in Section 5.

Theorem 8.9 (Triangular centroid identity)**.** *For every positive integer N,*

$$C_N = \frac{1}{N}\sum_{n=1}^{N} S_n = \frac{1}{N}\sum_{n=1}^{N}\sum_{k=1}^{n} e^{i\gamma_k} = \frac{1}{N}\sum_{k=1}^{N}(N-k+1)e^{i\gamma_k}. \tag{8.13}$$

Proof. The first equality is the definition (5.5). Replacing S_n by its defining sum gives the second equality, which is (5.6). In the double sum, the term $e^{i\gamma_k}$ appears exactly for those indices n with $k \le n \le N$. There are $N - k + 1$ such indices. Collecting equal terms gives the final expression, already recorded in (5.7). □

Corollary 8.10 (Triangular non-resonant centroid transform)**.** *The centroid C_N is a triangularly weighted non-resonant phase sum associated with the continuous frequency $X = e$.*

Proof. By the triangular centroid identity,

$$C_N = \frac{1}{N}\sum_{k=1}^{N}(N-k+1)e^{i\gamma_k}. \tag{8.14}$$

Thus C_N is a triangularly weighted sum of the phases $e^{i\gamma_k}$.

By the continuous-frequency identity at $X = e$, these phases coincide with the continuous-frequency phases $X^{i\gamma_k}$ evaluated at $X = e$. Since e is not contained in the arithmetic resonance set

$$\mathcal{R} = \{p^m : \ p \text{ prime}, \ m \in \mathbb{N}\}, \tag{8.15}$$

the triangularly weighted ordinal sum defining C_N is non-resonant in the Landau–Gonek prime-power sense.

□

8.5. **Equidistribution consequences.** Hlawka's theorem, applied with $\alpha = 1/(2\pi)$, implies the normalized cancellation already stated in (4.4). The next consequence records the same fact in the notation of this section.

Corollary 8.11 (Normalized cancellation at $X = e$)**.** *The ordinal phase sums at the continuous frequency $X = e$ satisfy*

$$\frac{S_N}{N} \longrightarrow 0 \qquad (N \to \infty). \tag{8.16}$$

Proof. Equation (4.4) is obtained from Hlawka's equidistribution theorem and Weyl's criterion with $\alpha = 1/(2\pi)$. Since $e^{i\gamma_k}$ is the phase associated with $X = e$ by Lemma 8.4, the result is exactly (8.16). □

Theorem 8.12 (Sublinear centroid growth under normalized cancellation)**.** *Assume that*

$$\frac{S_N}{N} \longrightarrow 0 \qquad (N \to \infty). \tag{8.17}$$

Then

$$\frac{C_N}{N} \longrightarrow 0 \qquad (N \to \infty). \tag{8.18}$$

Equivalently,

$$C_N = o(N). \tag{8.19}$$

Proof. By definition,

$$C_N = \frac{1}{N} \sum_{n=1}^{N} S_n. \tag{8.20}$$

Therefore

$$\frac{C_N}{N} = \frac{1}{N^2} \sum_{n=1}^{N} S_n. \tag{8.21}$$

Let $\varepsilon > 0$. Since

$$\frac{S_n}{n} \longrightarrow 0, \tag{8.22}$$

there exists $N_0 \in \mathbb{N}$ such that

$$|S_n| \leq \varepsilon n \tag{8.23}$$

for every $n \geq N_0$. For $N \geq N_0$, split the sum as

$$\frac{C_N}{N} = \frac{1}{N^2} \sum_{n=1}^{N_0-1} S_n + \frac{1}{N^2} \sum_{n=N_0}^{N} S_n. \tag{8.24}$$

Taking absolute values gives

$$\left|\frac{C_N}{N}\right| \leq \frac{1}{N^2} \sum_{n=1}^{N_0-1} |S_n| + \frac{1}{N^2} \sum_{n=N_0}^{N} |S_n|. \tag{8.25}$$

The first numerator is fixed, hence

$$\frac{1}{N^2} \sum_{n=1}^{N_0-1} |S_n| \longrightarrow 0. \tag{8.26}$$

For the second term, using (8.23), we obtain

$$\frac{1}{N^2} \sum_{n=N_0}^{N} |S_n| \leq \frac{\varepsilon}{N^2} \sum_{n=N_0}^{N} n \leq \frac{\varepsilon}{N^2} \sum_{n=1}^{N} n. \tag{8.27}$$

Since

$$\sum_{n=1}^{N} n = \frac{N(N+1)}{2}, \tag{8.28}$$

it follows that

$$\frac{\varepsilon}{N^2} \sum_{n=1}^{N} n = \frac{\varepsilon}{2} \left(1 + \frac{1}{N}\right). \tag{8.29}$$

Therefore

$$\limsup_{N\to\infty} \left|\frac{C_N}{N}\right| \leq \frac{\varepsilon}{2}. \tag{8.30}$$

Since $\varepsilon > 0$ is arbitrary, this proves (8.18), equivalently (8.19).

□

Corollary 8.13 (Equidistribution does not force centroid convergence)**.** *The normalized cancellation in Corollary 8.11 implies only*

$$C_N = o(N), \tag{8.31}$$

and does not imply

$$C_N \longrightarrow 0. \tag{8.32}$$

Proof. The sublinear estimate follows from Theorem 8.12. However, sublinear growth alone does not imply convergence to zero. For example, the scalar sequence

$$a_N = \sqrt{N} \tag{8.33}$$

satisfies

$$\frac{a_N}{N} \longrightarrow 0, \tag{8.34}$$

but does not converge to zero. Therefore normalized cancellation of the endpoint does not, by itself, force centroid convergence to zero. □

8.6. **Structural conclusion.** The preceding results separate the ordinal, height-truncated, and arithmetic layers of the construction. The sum S_N is an ordinal phase walk, while $S_X(T)$ is a height-truncated phase sum; the counting function $N_\zeta(T)$ provides the bridge between the two. The phase $e^{i\gamma}$ corresponds to the continuous frequency $X = e$. In the arithmetic Landau–Gonek formula the resonant frequencies are the integral prime powers. The present use of $X = e$ is a fixed positive real frequency.

The Chebyshev and logarithmic-derivative viewpoint explains the arithmetic origin of the resonant frequencies: the weights $\Lambda(x)$ arise from

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{\ell=1}^{\infty} \frac{\Lambda(\ell)}{\ell^s}, \tag{8.35}$$

and the Landau–Gonek main terms are supported on prime powers. With the extended real convention $\Lambda_{\mathbb{R}}(X) = 0$ away from prime powers, one has $\Lambda_{\mathbb{R}}(e) = 0$. Equivalently, e is non-resonant because it is not an integral prime power. Thus neither the phase walk S_N nor the triangular centroid C_N receives a direct Chebyshev–Landau main term. The centroid is a triangular non-resonant phase transform at the continuous frequency $X = e$.

This interpretation is compatible with Weyl and Hlawka equidistribution and with the Landau–Gonek prime-power resonance formula. Equidistribution gives normalized cancellation of the increments, and Theorem 8.12 gives the sublinear bound $C_N = o(N)$. Landau–Gonek identifies arithmetic resonances at prime powers, whereas the present construction uses the non-arithmetic frequency $X = e$. The conclusion is structural: it separates the ordinal and arithmetic layers and records the non-resonant role of the triangular centroid.

9. Conclusions

The paper isolates the following outputs.

9.1. **Complete-zero logarithmic confinement.**
Landau's fixed-real formula at $X = e$ gives a logarithmic confinement baseline for the complete-zero walk:

$$Z_N = O(\log N), \qquad \mathcal{C}_N = O(\log N), \qquad \mathcal{M}_0(N) = O\Big((\log N)^2\Big). \tag{9.1}$$

Under RH this specializes to the original unit phase walk through

$$Z_N = S_N, \qquad \mathcal{C}_N = C_N, \qquad \mathcal{M}_0(N) = M_0(N). \tag{9.2}$$

9.2. **Exact centered pair statistic.**
In the complete-zero setting,

$$\mathcal{M}_0(N) = \sum_{j,k=1}^{N} W_N(j,k) e^{\rho_j - \frac{1}{2}} e^{\overline{\rho_k} - \frac{1}{2}}. \tag{9.3}$$

Under RH, this becomes the exact centered pair statistic

$$M_0(N) = \sum_{j,k=1}^{N} W_N(j,k) e^{i(\gamma_j - \gamma_k)} \tag{9.4}$$

Here W_N is the exact centering kernel

$$W_N(j,k) = \frac{N - \max(j,k) + 1}{N} - \frac{(N-j+1)(N-k+1)}{N^2}. \tag{9.5}$$

It is the weight forced by subtracting the triangular centroid before forming the quadratic moment.

9.3. **Brownian-bridge/Green kernel.**
After reversing indices, the kernel becomes

$$K_N(r,s) = \frac{\min(r,s)}{N} - \frac{rs}{N^2}, \tag{9.6}$$

the discrete Brownian-bridge covariance kernel, equivalently the Green kernel of the discrete Dirichlet Laplacian. This is the central structural point of the paper.

9.4. **Lifted phases and Green-modal coordinates.**
There is no modal shortcut hidden in the notation. The lifted phase representation in Lemma 6.14 separates the angular representative θ_k from the winding number B_k. For $t = 1$, the winding is invisible in $e^{i\gamma_k}$; for real frequency deformations $e^{it\gamma_k}$, and in particular in the Montgomery scaling, the winding data are part of the phase.

The modal basis is the normalized Dirichlet sine basis (6.97). Its orthonormality is proved in Lemma 6.15, and exact modal inversion is Lemma 6.19. Thus the coefficient vector $A(N;t)$ is genuine coordinate data, not a heuristic compression.

With these coordinates, the Green kernel diagonalizes and yields

$$\mathcal{E}_N(t) = A(N;t)^* \Lambda_N A(N;t) = \sum_{m=1}^{N-1} \lambda_m |A_m(N;t)|^2. \tag{9.7}$$

Under RH at $t = 1$, this is the centered phase-walk moment $M_0(N)$.

9.5. **Exact recovery versus scalar energy bounds.**
Put

$$S_N(t) = \sum_{k=1}^{N} e^{it\gamma_k}. \tag{9.8}$$

The endpoint is recovered exactly from the Green-modal coefficient data:

$$S_N(t) = e^{it\gamma_1} + \sum_{\substack{1 \le m \le N-1 \\ m \text{ odd}}} u_m A_m(N;t). \tag{9.9}$$

Even modes vanish from the endpoint sum only because their projection onto the constant vector is zero; they still contribute to the centered Green energy.

The scalar Green energy alone does not recover the endpoint or the centroid: it discards modal direction and phase information. Exact recovery uses the coefficient vector $A(N;t)$, equivalently the weighted modal data $\Lambda_N A(N;t)$, as in (6.203). The scalar energy gives only one-sided deweighting estimates. From (6.186) and (6.209),

$$\left|S_N(t) - e^{it\gamma_1}\right|^2 \le 2N\,\mathcal{E}_N(t), \tag{9.10}$$

and

$$\left|C_N(t) - e^{it\gamma_1}\right|^2 \le (N-1)\mathcal{E}_N(t). \tag{9.11}$$

Consequently, if $\mathcal{E}_N(1) = O((\log N)^2)$, the Green bounds give only the weaker estimates

$$S_N = O(\sqrt{N}\log N), \qquad C_N = O(\sqrt{N}\log N). \tag{9.12}$$

The logarithmic bounds in (9.1) are stronger and come from Landau's fixed-real formula at $X = e$, not from the scalar Green-energy inequality.

9.6. **Green-modal Montgomery formulation.**
The smoothed Montgomery form factor can be written as

$$F_N(\tau) = \frac{1}{N}\int_{-\infty}^{\infty} e^{-2|y|}\left|e^{i(\tau\log(T_N/2\pi)+y)\gamma_1} + \sum_{\substack{1\le m\le N-1\\ m\ \text{odd}}} u_m A_m(N;\tau\log(T_N/2\pi)+y)\right|^2 dy. \tag{9.13}$$

Thus the Montgomery–GUE form-factor target can be expressed as a statement about the Green-modal endpoint projection:

$$F_N(\tau) \to \min(|\tau|, 1). \tag{9.14}$$

The finite modal bounds on individual $A_m(N;t)$ are unconditional, but they do not determine this limit. The conditional modal covariance criterion is stated in Corollary 6.34: the limiting form factor is governed by the rank-one contraction $N^{-1}u^*\Sigma_N(\tau)u$, not by pointwise modal bounds alone.

The final logical division is therefore sharp: Landau gives the strong logarithmic confinement estimates; the Green kernel gives the exact structural identity, the modal decomposition, and the exact deweighting of linear functionals. The scalar Green energy supplies canonical upper bounds, not exact recovery of endpoint-type quantities.

9.7. **RVM residual decomposition.**
Here RVM means Riemann–von Mangoldt. Let $\widetilde{\gamma}_k = N_0^{-1}(k)$ denote the synthetic ordinates obtained by inverting the smooth Riemann–von Mangoldt counting main term. These ordinates form the rigid RVM background against which the true zeta ordinates γ_k are compared, and write $d_k = \gamma_k - \widetilde{\gamma}_k$ for the residual displacement.

The modes split as

$$A_m(N;t) = A_m^{\mathrm{RVM}}(N;t) + R_m(N;t), \tag{9.15}$$

with

$$\sum_{m=1}^{N-1} |R_m(N;t)|^2 = \sum_{k=2}^{N} |e^{itd_k} - 1|^2 \le 4(N-1). \tag{9.16}$$

Equivalently,

$$\frac{1}{N}\sum_{m=1}^{N-1}|R_m(N;t)|^2 \le 4. \tag{9.17}$$

Thus $A_m^{\mathrm{RVM}}(N;t)$ is the modal coefficient computed from the synthetic RVM ordinates $\widetilde{\gamma}_k$, while $R_m(N;t)$ is the residual modal difference contributed by replacing those synthetic ordinates with the true zero ordinates. In other words, R_m measures the mode-by-mode remainder between the synthetic RVM grid and the actual zeros.

The main contribution of the paper is the identification of the Brownian-bridge/Green kernel hidden in the centered phase-walk geometry. This kernel converts the centroid problem into an exact weighted pair-correlation statistic and provides a natural Green-modal language in which the Montgomery–GUE form factor can be expressed. The centered moment gives an exact Green-kernel weighted pair statistic. The Green-modal coordinates give an exact endpoint representation of the Montgomery form factor, while the scalar Green energy gives the corresponding one-sided deweighting estimate.

9.8. **Structural separation.**

The fixed-frequency objects $S_N(1)$, C_N, and $M_0(N)$ describe non-resonant Landau cancellation and Green-weighted centered geometry. The Montgomery–GUE statistics are seen through the deformed endpoint family $S_N(t)$. In Fourier variables this gives the form factor, while in the unfolded pair variable

$$\xi_{jk} = \frac{\log(T_0/2\pi)}{2\pi}(\gamma_j - \gamma_k) \tag{9.18}$$

it gives the sine-kernel pair correlation. The Green-modal vector $A(N;t)$ is the common coordinate system linking these two regimes.

References

[1] B. Riemann, *Über die Anzahl der Primzahlen unter einer gegebenen Grösse*, Monatsberichte der Berliner Akademie, 1859, 671–680.
Reprinted in:
B. Riemann, *Gesammelte Mathematische Werke*, Dover Publications, 1953.

[2] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, Second Edition, revised by D. R. Heath-Brown, Oxford University Press, 1986.

[3] H. M. Edwards, *Riemann's Zeta Function*, Dover Publications, 2001.

[4] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, American Mathematical Society, 2004.

[5] H. Weyl, *Über die Gleichverteilung von Zahlen mod. Eins*, Mathematische Annalen **77** (1916), 313–352. doi:10.1007/BF01475864.

[6] H. Rademacher, *On the distribution of the values of the zeta function*, Colloquium Mathematicum **4** (1956), 79–81.

[7] E. Hlawka, *Über die Gleichverteilung gewisser Folgen, die mit den Nullstellen der Zetafunktion zusammenhängen*, Sitzungsberichte der Österreichischen Akademie der Wissenschaften, Mathematisch-Naturwissenschaftliche Klasse, Abteilung II **184** (1975), 459–471.

[8] E. Landau, *Über die Nullstellen der Zetafunktion*, Mathematische Annalen **71** (1911), 548–564. doi:10.1007/BF01456881.

[9] S. M. Gonek, *A formula of Landau and mean values of $\zeta(s)$*, in: *Topics in Analytic Number Theory*, edited by S. W. Graham and J. D. Vaaler, University of Texas Press, Austin, 1985, 92–97.

[10] S. M. Gonek, *An explicit formula of Landau and its applications to the theory of the zeta-function*, in: *A Tribute to Emil Grosswald: Number Theory and Related Analysis*, Contemporary Mathematics **143**, American Mathematical Society, 1993, 395–413.

[11] A. M. Odlyzko, *On the distribution of spacings between zeros of the zeta function*, Mathematics of Computation **48** (1987), 273–308. doi:10.1090/S0025-5718-1987-0866115-0.

[12] J. Kaczorowski, A. Languasco, A. Perelli, *A note on Landau's formula*, Functiones et Approximatio Commentarii Mathematici **28** (2000), 173–186. doi:10.7169/facm/1538186693.

[13] H. L. Montgomery, *The pair correlation of zeros of the zeta function*, in: *Analytic Number Theory*, Proceedings of Symposia in Pure Mathematics **24**, American Mathematical Society, 1973, 181–193.
[14] P. Borwein, S. Choi, B. Rooney, A. Weirathmueller, *The Riemann Hypothesis: A Resource for the Afficionado and Virtuoso Alike*, Springer, 2008.
[15] K. Ford and A. Zaharescu, *Unnormalized Differences Between Zeros of the Riemann Zeta Function*, Acta Arithmetica **117** (2005), 163–176. doi:10.4064/aa117-2-5.
[16] K. Ford, K. Soundararajan, A. Zaharescu, *The Distribution of the Ordinates of the Zeros of the Riemann Zeta Function*, arXiv:2407.14173, 2024.
[17] A. Fujii, *On the uniformity of the distribution of the zeros of the Riemann zeta function*, Journal für die reine und angewandte Mathematik **302** (1978), 167–205.

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